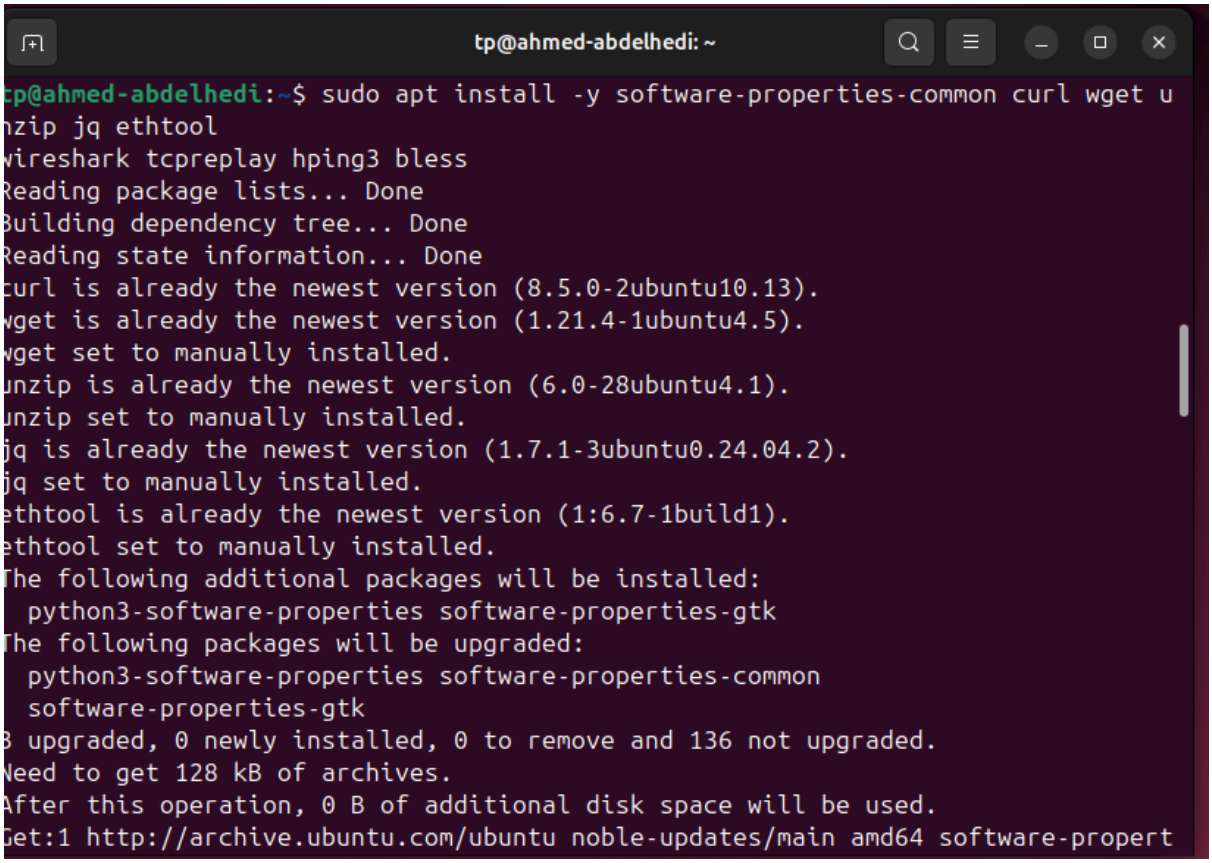
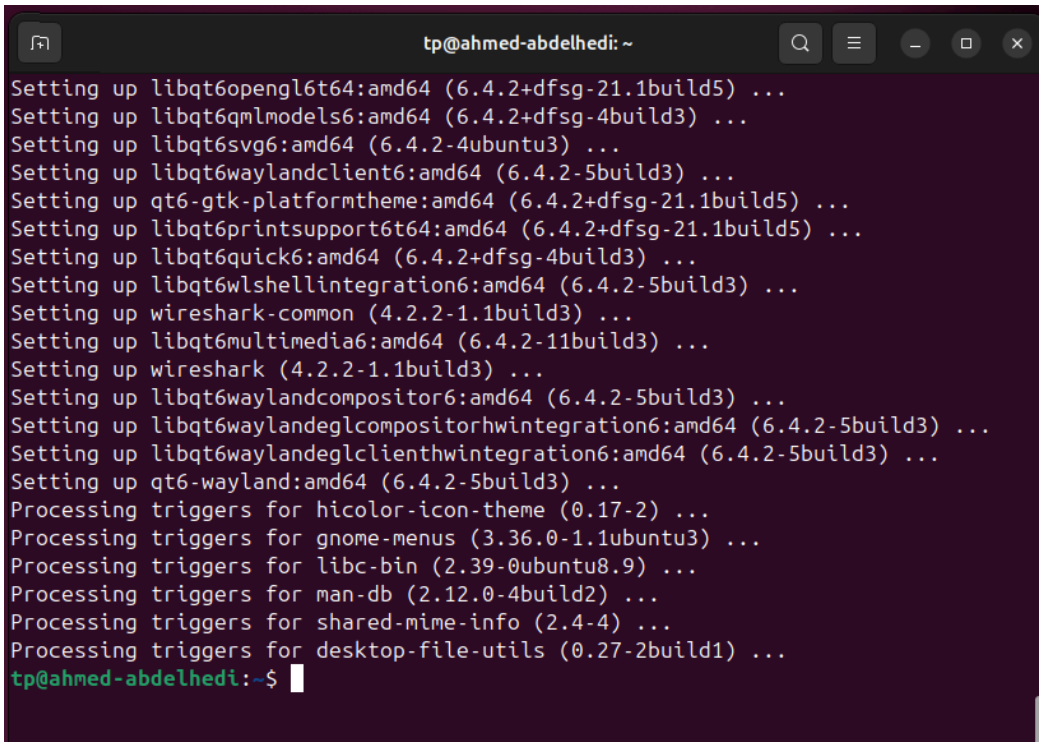
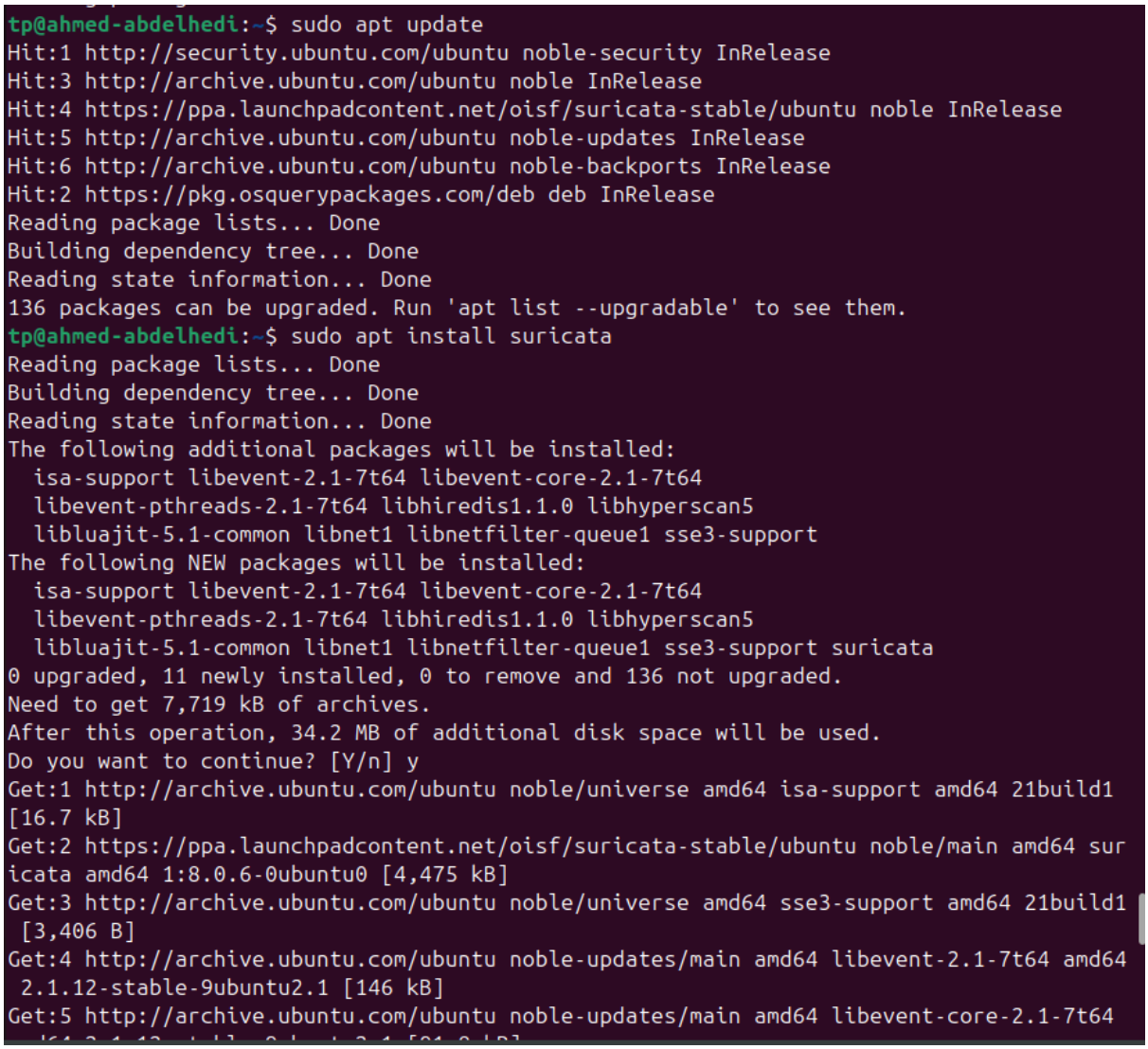
Lab 2 : Network Intrusion Detection using Suricata

Ahmed Abdelhedi

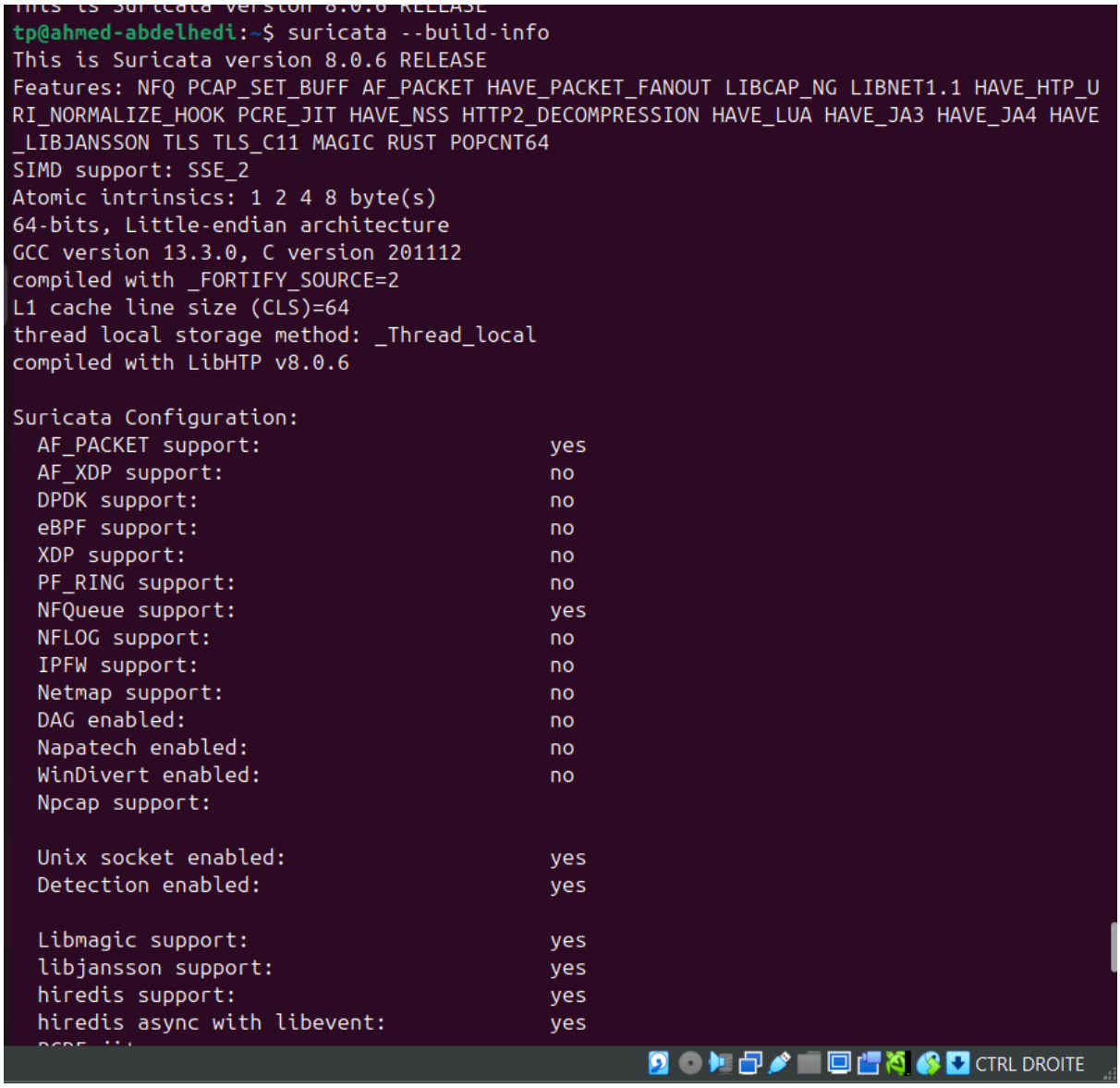




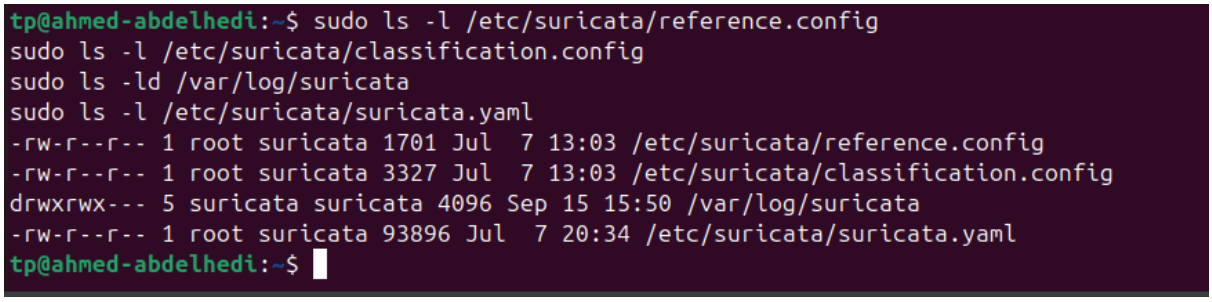
3:


tp@ahmed-abdelhedi:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:4 https://ppa.launchpadcontent.net/oisf/suricata-stable/ubuntu noble InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:2 https://pkg.osquerypackages.com/deb deb InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
136 packages can be upgraded. Run 'apt list --upgradable' to see them.
tp@ahmed-abdelhedi:~$ sudo apt install suricata
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
isa-support libevent-2.1-7t64 libevent-core-2.1-7t64
libevent-pthreads-2.1-7t64 libhiredis1.1.0 libhyperscan5
libluajit-5.1-common libnet1 libnetfilter-queue1 sse3-support
The following NEW packages will be installed:
isa-support libevent-2.1-7t64 libevent-core-2.1-7t64
libevent-pthreads-2.1-7t64 libhiredis1.1.0 libhyperscan5
libluajit-5.1-common libnet1 libnetfilter-queue1 sse3-support suricata
0 upgraded, 11 newly installed, 0 to remove and 136 not upgraded.
Need to get 7,719 kB of archives.
After this operation, 34.2 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 isa-support amd64 21build1
[16.7 kB]
Get:2 https://ppa.launchpadcontent.net/oisf/suricata-stable/ubuntu noble/main amd64 sur
icata amd64 1:8.0.6-0ubuntu0 [4,475 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble/universe amd64 sse3-support amd64 21build1
[3,406 B]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libevent-2.1-7t64 amd64
2.1.12-stable-9ubuntu2.1 [146 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libevent-core-2.1-7t64

5:


tp@ahmed-abdelhedi:~$ suricata --build-info
This is Suricata version 8.0.6 RELEASE
Features: NFQ PCAP_SET_BUFF AF_PACKET HAVE_PACKET_FANOUT LIBCAP_NG LIBNET1.1 HAVE_HTP_URI_NORMALIZE_HOOK PCRE_JIT HAVE_NSS HTTP2_DECOMPRESSION HAVE_LUA HAVE_JA3 HAVE_JA4 HAVE_LIBJANSSON TLS TLS_C11 MAGIC RUST POPCNT64
SIMD support: SSE_2
Atomic intrinsics: 1 2 4 8 byte(s)
64-bits, Little-endian architecture
GCC version 13.3.0, C version 201112
compiled with _FORTIFY_SOURCE=2
L1 cache line size (CLS)=64
thread local storage method: _Thread_local
compiled with LibHTP v8.0.6
Suricata Configuration:
AF_PACKET support: yes
AF_XDP support: no
DPDK support: no
eBPF support: no
XDP support: no
PF_RING support: no
NFQueue support: yes
NFLOG support: no
IPFW support: no
Netmap support: no
DAG enabled: no
Napatech enabled: no
WinDivert enabled: no
Npcap support:
Unix socket enabled: yes
Detection enabled: yes
Libmagic support: yes
libjansson support: yes
hiredis support: yes
hiredis async with libevent: yes
CTRL DROITE

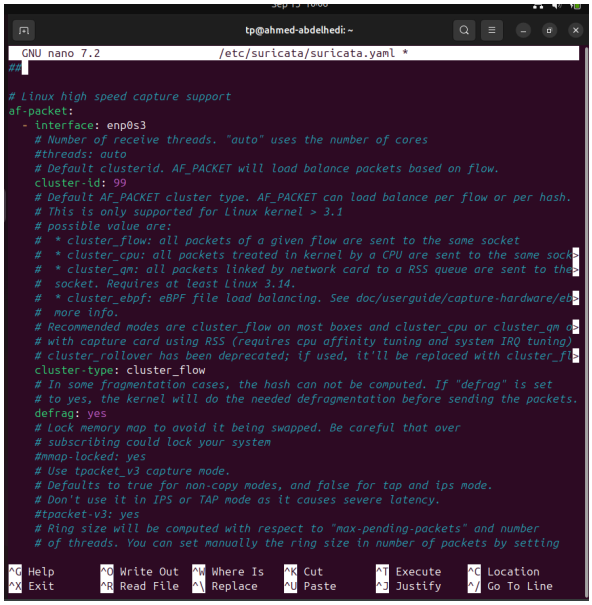
6:

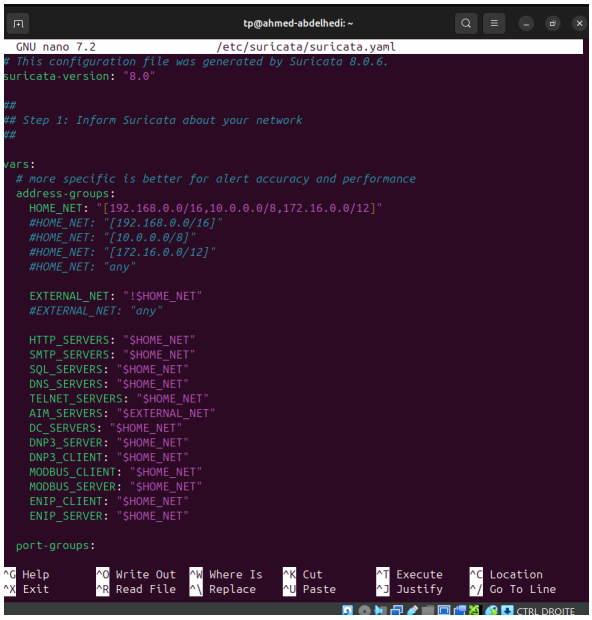

tp@ahmed-abdelhedi:~$ sudo ls -l /etc/suricata/reference.config
sudo ls -l /etc/suricata/classification.config
sudo ls -ld /var/log/suricata
sudo ls -l /etc/suricata/suricata.yaml
-rw-r--r-- 1 root suricata 1701 Jul 7 13:03 /etc/suricata/reference.config
-rw-r--r-- 1 root suricata 3327 Jul 7 13:03 /etc/suricata/classification.config
drwxrwx--- 5 suricata suricata 4096 Sep 15 15:50 /var/log/suricata
-rw-r--r-- 1 root suricata 93896 Jul 7 20:34 /etc/suricata/suricata.yaml
tp@ahmed-abdelhedi:~$

III. Suricata Configuration and Network Setup:

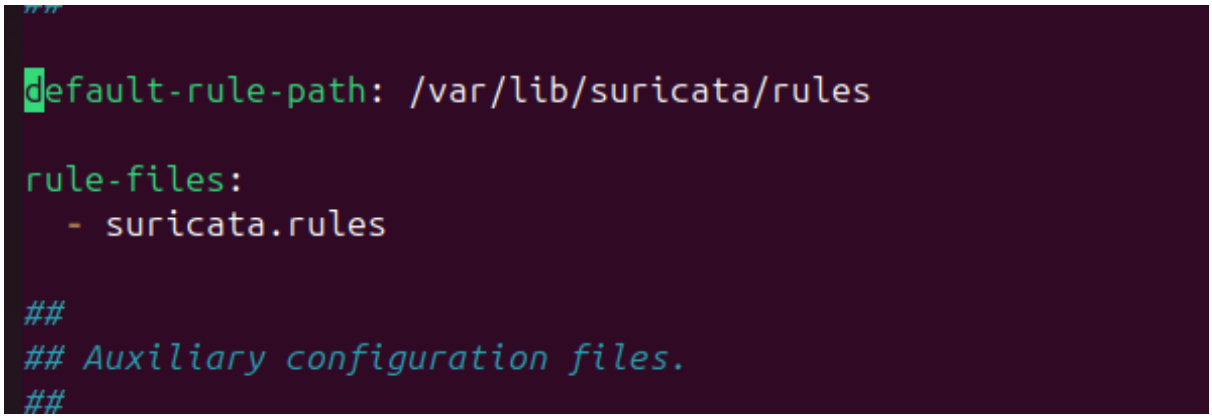
7:

```
tp@ahmed-abdelhedi:~$ sudo cp /etc/suricata/suricata.yaml /etc/suricata/suricata.yaml.b
ackup
tp@ahmed-abdelhedi:~$ sudo ls -l /etc/suricata/suricata.yaml.backup
-rw-r--r-- 1 root root 93896 Sep 15 16:00 /etc/suricata/suricata.yaml.backup
tp@ahmed-abdelhedi:~$ ip -br addr
lo               UNKNOWN        127.0.0.1/8 ::1/128
enp0s3           UP             10.0.2.15/24 fd00::6de8:276a:d989:e6f0/64 fd00::bc74:2e
21:7ad7:9393/64 fe80::8fb4:b9d1:81a:bb2c/64
tp@ahmed-abdelhedi:~$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
tp@ahmed-abdelhedi:~$
```





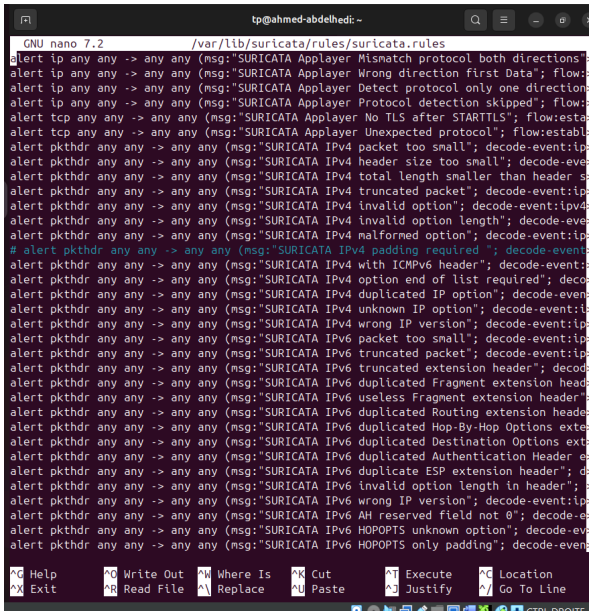
Check also that the default values of the variables HTTP_SERVERS, SMTP_SERVERS, SQL_SERVERS, DNS_SERVERS and TELNET_SERVERS, are set to HOME_NET :
yes ,they are set to HOME_NET



12:

```
tp@ahmed-abdelhedi:~$ sudo nano /etc/suricata/suricata.yaml
tp@ahmed-abdelhedi:~$ sudo suricata-update
15/9/2026 -- 16:15:52 - <Info> -- Using data-directory /var/lib/suricata.
15/9/2026 -- 16:15:52 - <Info> -- Using Suricata configuration /etc/suricata/suricata.y
aml
15/9/2026 -- 16:15:52 - <Info> -- Using /usr/share/suricata/rules for Suricata provided
 rules.
15/9/2026 -- 16:15:52 - <Info> -- Found Suricata version 8.0.6 at /usr/bin/suricata.
15/9/2026 -- 16:15:52 - <Info> -- Loading /etc/suricata/suricata.yaml
15/9/2026 -- 16:15:52 - <Info> -- Disabling rules for protocol pgsql
15/9/2026 -- 16:15:52 - <Info> -- Disabling rules for protocol modbus
15/9/2026 -- 16:15:52 - <Info> -- Disabling rules for protocol dnp3
15/9/2026 -- 16:15:52 - <Info> -- Disabling rules for protocol enip
15/9/2026 -- 16:15:52 - <Info> -- No sources configured, will use Emerging Threats Open
15/9/2026 -- 16:15:52 - <Info> -- Fetching https://rules.emergingthreats.net/open/suric
ata-8.0.6/emerging.rules.tar.gz.
  12% - 679936/5608418
```

13:

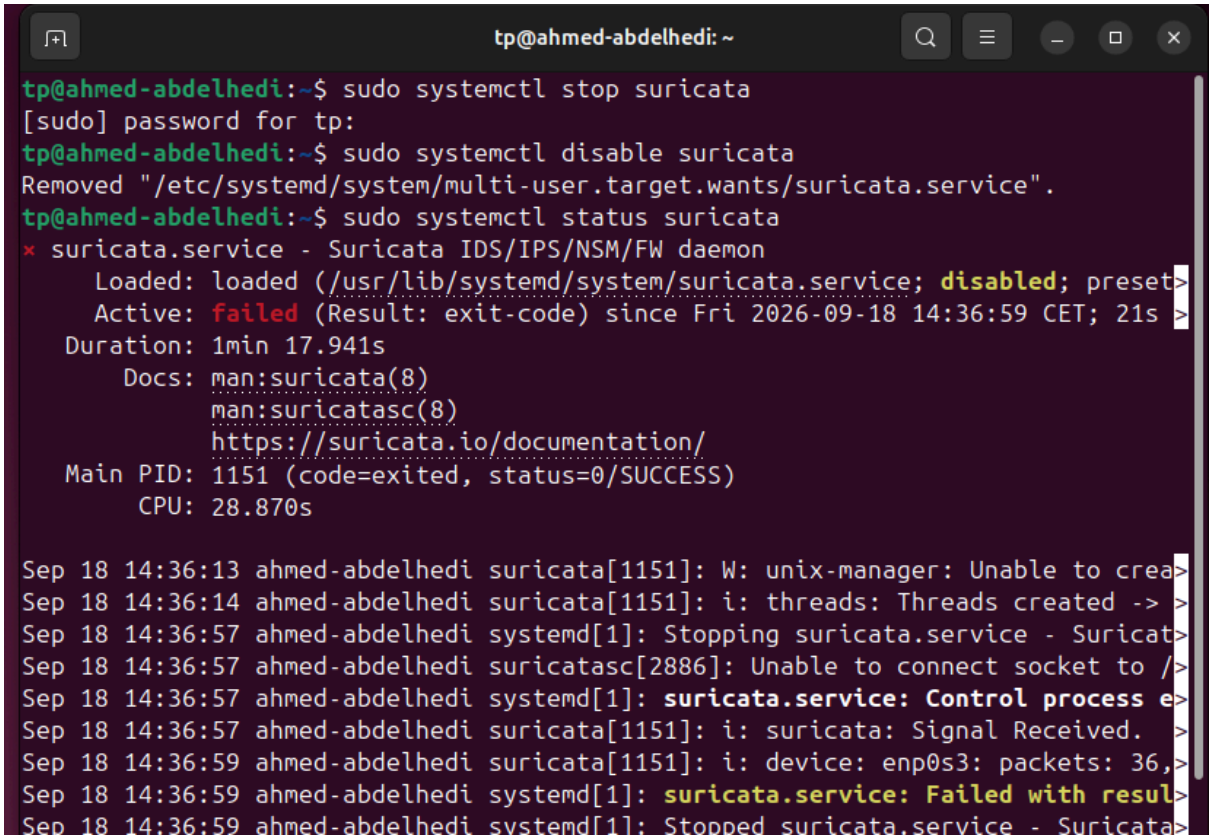


14:

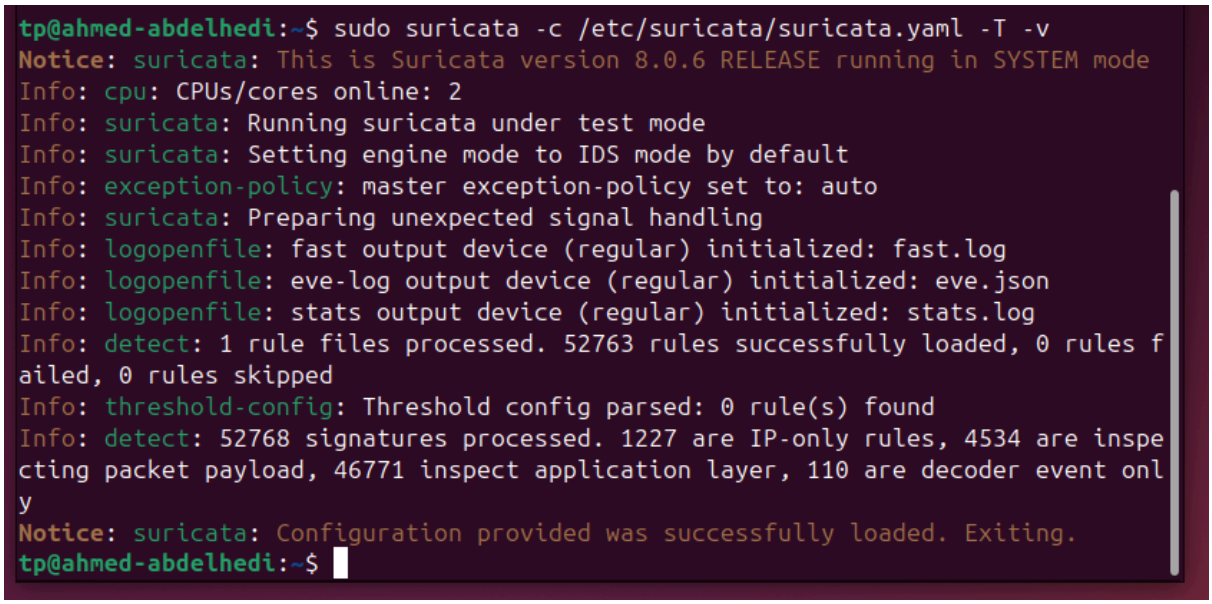
```
tp@ahmed-abdelhedi:~$ sudo wc -l /var/lib/suricata/rules/suricata.rules
68716 /var/lib/suricata/rules/suricata.rules
tp@ahmed-abdelhedi:~$
```

there are 68716 rules

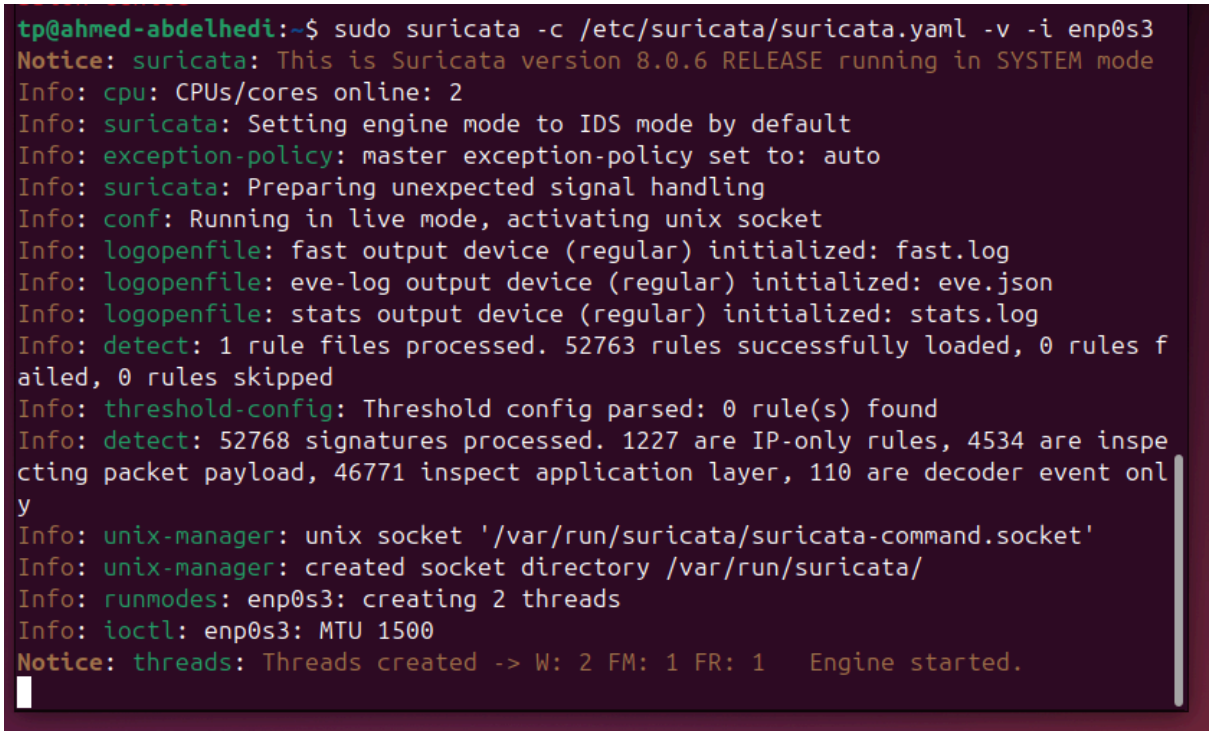
15:


tp@ahmed-abdelhedi: ~
tp@ahmed-abdelhedi:~$ sudo systemctl stop suricata
[sudo] password for tp:
tp@ahmed-abdelhedi:~$ sudo systemctl disable suricata
Removed "/etc/systemd/system/multi-user.target.wants/suricata.service".
tp@ahmed-abdelhedi:~$ sudo systemctl status suricata
× suricata.service - Suricata IDS/IPS/NSM/FW daemon
Loaded: loaded (/usr/lib/systemd/system/suricata.service; disabled; preset
Active: failed (Result: exit-code) since Fri 2026-09-18 14:36:59 CET; 21s
Duration: 1min 17.941s
Docs: man:suricata(8)
man:suricatasc(8)
https://suricata.io/documentation/
Main PID: 1151 (code=exited, status=0/SUCCESS)
CPU: 28.870s
Sep 18 14:36:13 ahmed-abdelhedi suricata[1151]: W: unix-manager: Unable to crea
Sep 18 14:36:14 ahmed-abdelhedi suricata[1151]: i: threads: Threads created ->
Sep 18 14:36:57 ahmed-abdelhedi systemd[1]: Stopping suricata.service - Suricat
Sep 18 14:36:57 ahmed-abdelhedi suricatasc[2886]: Unable to connect socket to /
Sep 18 14:36:57 ahmed-abdelhedi systemd[1]: suricata.service: Control process e
Sep 18 14:36:57 ahmed-abdelhedi suricata[1151]: i: suricata: Signal Received.
Sep 18 14:36:59 ahmed-abdelhedi suricata[1151]: i: device: enp0s3: packets: 36,
Sep 18 14:36:59 ahmed-abdelhedi systemd[1]: suricata.service: Failed with resul

16:


tp@ahmed-abdelhedi:~$ sudo suricata -c /etc/suricata/suricata.yaml -T -v
Notice: suricata: This is Suricata version 8.0.6 RELEASE running in SYSTEM mode
Info: cpu: CPUs/cores online: 2
Info: suricata: Running suricata under test mode
Info: suricata: Setting engine mode to IDS mode by default
Info: exception-policy: master exception-policy set to: auto
Info: suricata: Preparing unexpected signal handling
Info: logopenfile: fast output device (regular) initialized: fast.log
Info: logopenfile: eve-log output device (regular) initialized: eve.json
Info: logopenfile: stats output device (regular) initialized: stats.log
Info: detect: 1 rule files processed. 52763 rules successfully loaded, 0 rules failed, 0 rules skipped
Info: threshold-config: Threshold config parsed: 0 rule(s) found
Info: detect: 52768 signatures processed. 1227 are IP-only rules, 4534 are inspecting packet payload, 46771 inspect application layer, 110 are decoder event only
Notice: suricata: Configuration provided was successfully loaded. Exiting.
tp@ahmed-abdelhedi:~$

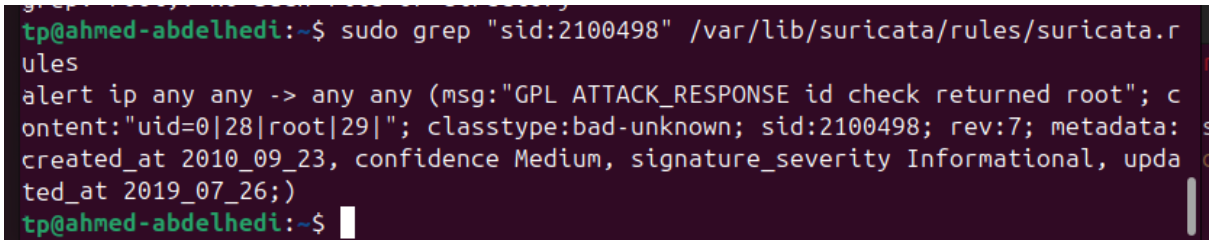
17:



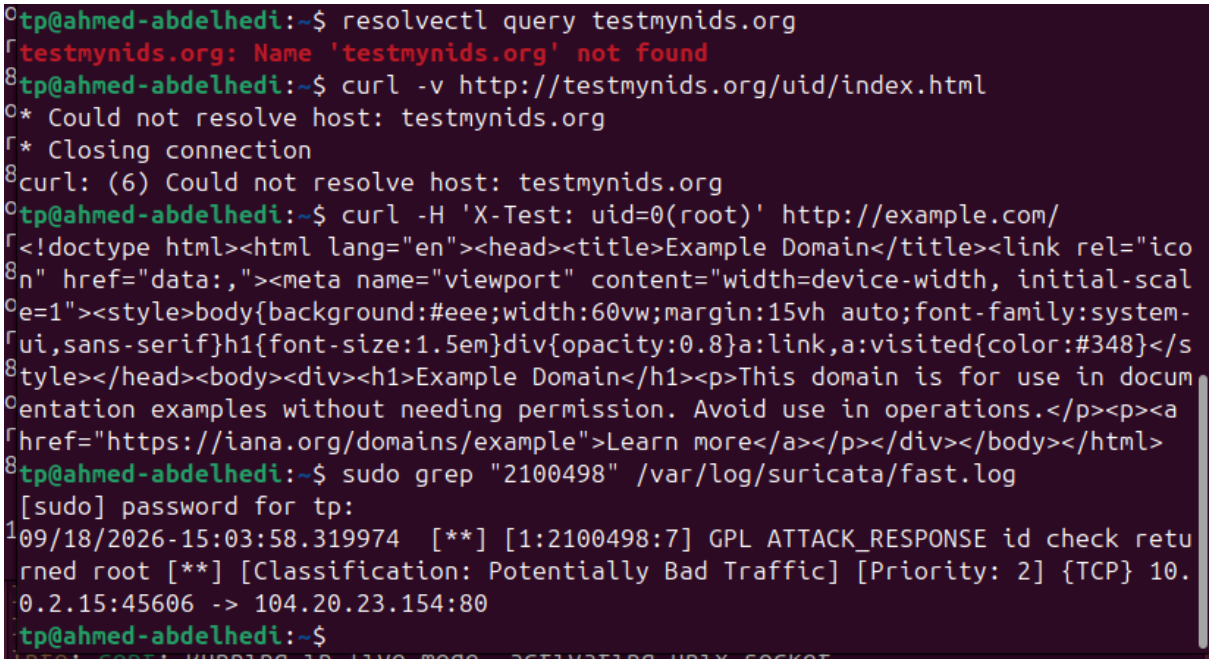
IV. Testing Suricata's Built-In Rules:

IV.1. Detection of Root-Level Compromise:

18:



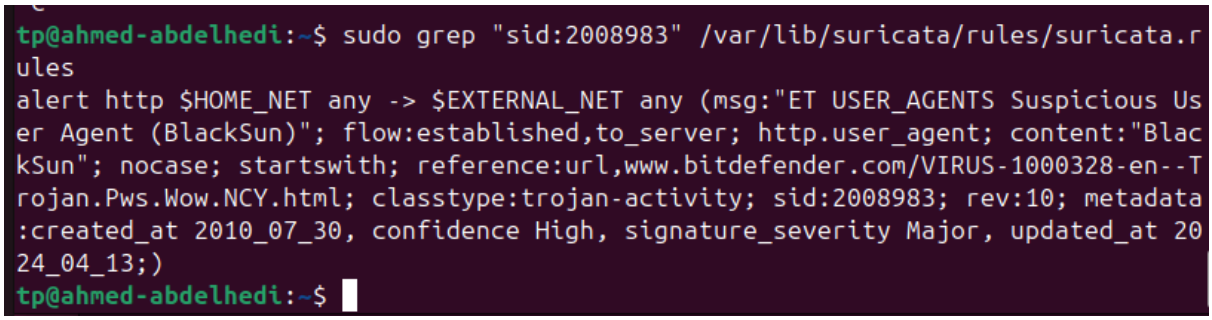
19:



Since the test domain was unavailable, we generated equivalent HTTP traffic containing uid=0(root) to verify that Suricata could detect and log the event.

IV.2. Detection of Malware User-Agents (BlackSun):

20:



flow:to_server,established → The rule applies to an established connection with traffic going to the server.

http.user_agent → Inspects the HTTP User-Agent field.

content:"BlackSun" → Searches for the string BlackSun.

nocase → Makes the search case-insensitive.

depth:8 → Limits the search to the first 8 bytes of the inspected content.

startswith → Requires the string BlackSun to appear at the beginning of the inspected HTTP User-Agent field

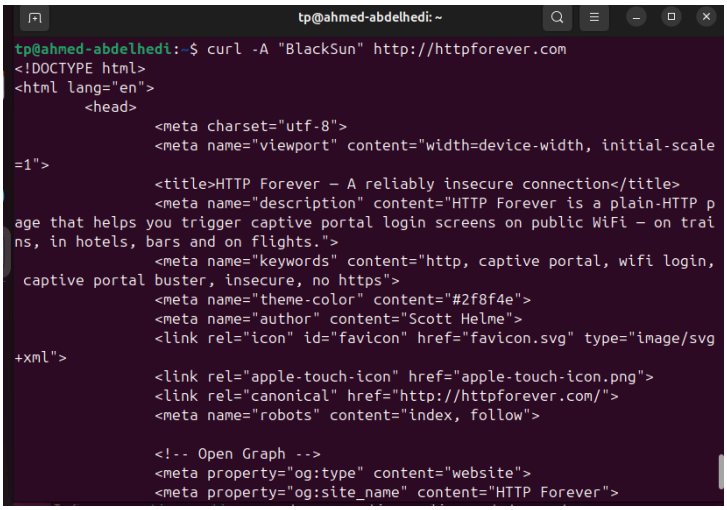
reference → Provides an external reference related to the detected threat.

classtype → Defines the category of the detected attack or activity.

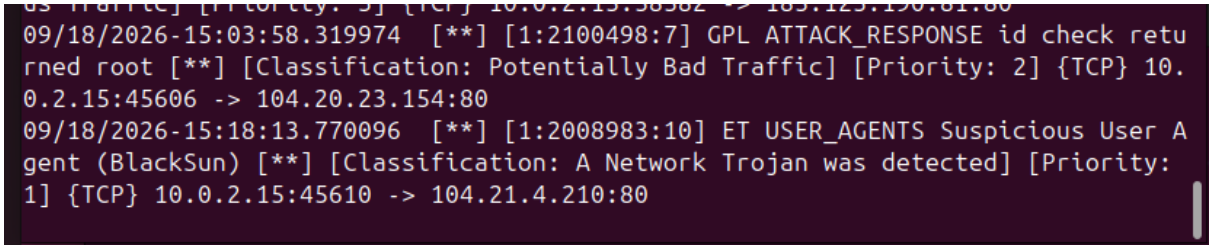
sid → Unique identifier of the Suricata rule.

rev → Indicates the revision/version number of the rule.

21:



22:



23:

```
tp@ahmed-abdelhedi:~$ curl -A "BlackSunBLABLABLA" http://httpforever.com
<!DOCTYPE html>
<html lang="en">
        <head>
                <meta charset="utf-8">
                <meta name="viewport" content="width=device-width, initial-scale
=1">
```

```
tp@ahmed-abdelhedi:~$ curl -A "BLABLABLABlackSun" http://httpforever.com
<!DOCTYPE html>
<html lang="en">
        <head>
                <meta charset="utf-8">
                <meta name="viewport" content="width=device-width, initial-scale
=1">
```

```
09/18/2026-15:03:58.319974  [**] [1:2100498:7] GPL ATTACK_RESPONSE id check retu
rned root [**] [Classification: Potentially Bad Traffic] [Priority: 2] {TCP} 10.
0.2.15:45606 -> 104.20.23.154:80
09/18/2026-15:18:13.770096  [**] [1:2008983:10] ET USER_AGENTS Suspicious User A
gent (BlackSun) [**] [Classification: A Network Trojan was detected] [Priority:
1] {TCP} 10.0.2.15:45610 -> 104.21.4.210:80
09/18/2026-15:24:46.062314  [**] [1:2008983:10] ET USER_AGENTS Suspicious User A
gent (BlackSun) [**] [Classification: A Network Trojan was detected] [Priority:
1] {TCP} 10.0.2.15:38136 -> 104.21.4.210:80
```

The first command triggers an alert because BlackSun is at the beginning of the User-Agent. The second command does not trigger an alert because BlackSun appears later. This is due to the startswith condition in the Suricata rule.

IV.3. Detection of Malicious IP Sources (Spamhaus DROP):

24:

```
tp@ahmed-abdelhedi:~$ sudo grep "sid:2400000" /var/lib/suricata/rules/suricata.r
ules
alert ip [2.56.192.0/22,2.57.17.0/24,2.57.122.0/24,2.57.232.0/23,2.57.234.0/23,2
.58.56.0/24,2.59.152.0/24,2.59.153.0/24,5.42.92.0/24,5.101.86.0/24,5.105.220.0/2
4,5.175.169.0/24,5.175.189.0/24,5.183.60.0/22,5.188.236.0/23,5.230.201.0/24,5.23
1.63.0/24,5.252.83.0/24,5.252.153.0/24,5.252.155.0/24] any -> $HOME_NET any (msg
:"ET DROP Spamhaus DROP Listed Traffic Inbound group 1"; reference:url,www.spamh
aus.org/drop/drop.txt; threshold: type limit, track by_src, seconds 3600, count
1; classtype:misc-attack; flowbits:set,ET.Evil; flowbits:set,ET.DROPIP; sid:2400
000; rev:4830; metadata:affected_product Any, attack_target Any, deployment Peri
meter, tag Dshield, signature_severity Minor, created_at 2010_12_30, updated_at
2026_09_14;)
tp@ahmed-abdelhedi:~$
```

25:

type limit → limits how often the alert is generated.

track by_src → the limit is applied separately for each source IP.

seconds 3600 → the time window is 3600 seconds = 1 hour.

count 1 → only one alert is generated per source IP during that hour.

So, The rule generates at most one alert per source IP every hour.

26:

```
tp@ahmed-abdelhedi:~$ sudo apt install hping3
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  hping3
0 upgraded, 1 newly installed, 0 to remove and 149 not upgraded.
Need to get 100.0 kB of archives.
After this operation, 237 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 hping3 amd64 3.a2.ds
2-10build2 [100.0 kB]
Fetched 100.0 kB in 1s (138 kB/s)
Selecting previously unselected package hping3.
(Reading database ... 197247 files and directories currently installed.)
Preparing to unpack .../hping3_3.a2.ds2-10build2_amd64.deb ...
Unpacking hping3 (3.a2.ds2-10build2) ...
Setting up hping3 (3.a2.ds2-10build2) ...
Processing triggers for man-db (2.12.0-4build2) ...
tp@ahmed-abdelhedi:~$
```

27:

```
tp@ahmed-abdelhedi:~$ sudo hping3 -S -a 2.57.234.1 -p 2020 -c 5 10.0.2.99
HPING 10.0.2.99 (enp0s3 10.0.2.99): S set, 40 headers + 0 data bytes

--- 10.0.2.99 hping statistic ---
5 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```

28:

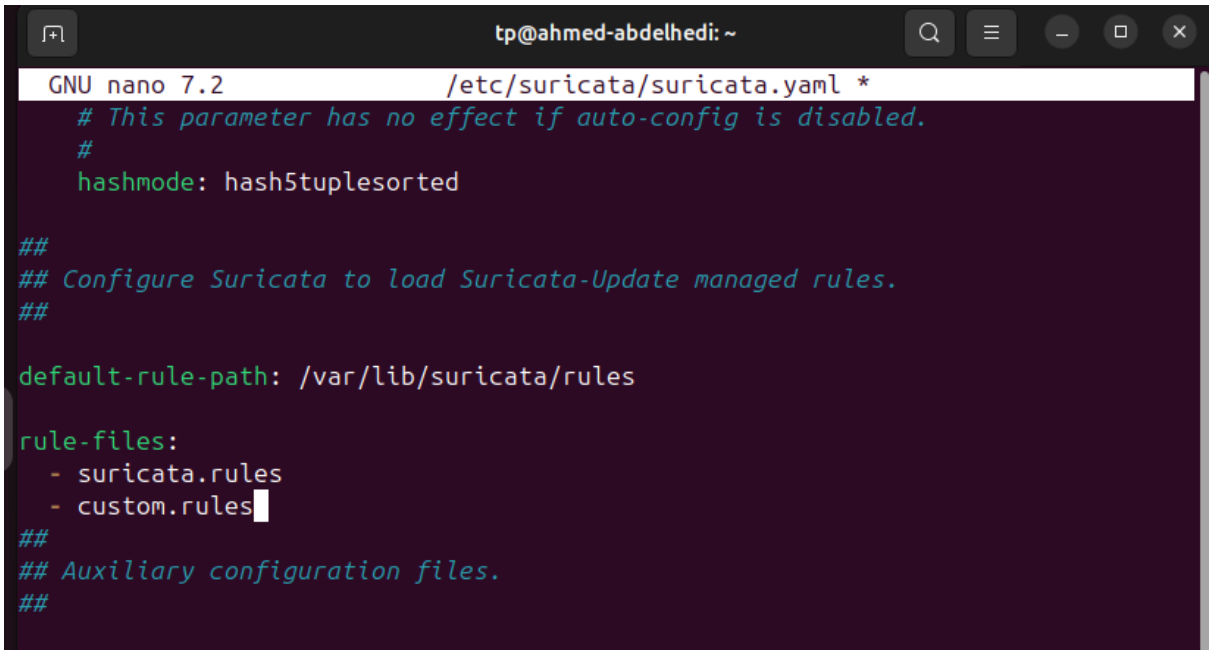
```
09/18/2026-15:51:13.764085  [**] [1:2400000:4830] ET DROP Spamhaus DROP Listed T
raffic Inbound group 1 [**] [Classification: Misc Attack] [Priority: 2] {TCP} 2.
57.234.1:1624 -> 10.0.2.99:2020
tp@ahmed-abdelhedi:~$
```

V. Adding our own rules to Suricata and generating alerts :

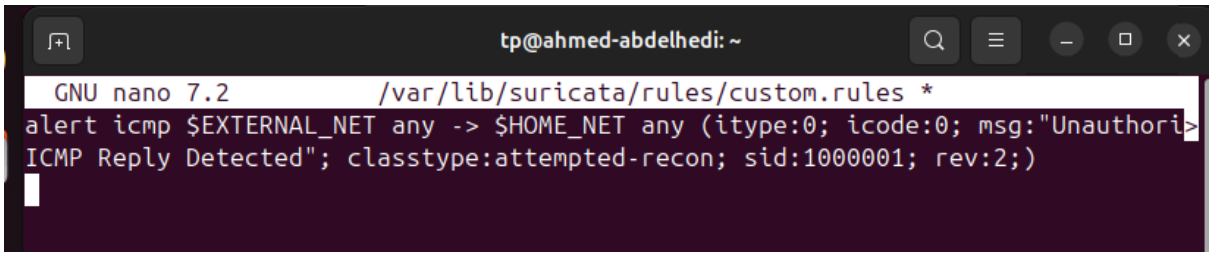
29:

```
tp@ahmed-abdelhedi:~$ sudo touch /var/lib/suricata/rules/custom.rules
[sudo] password for tp:
tp@ahmed-abdelhedi:~$ ls -l /var/lib/suricata/rules/custom.rules
ls: cannot access '/var/lib/suricata/rules/custom.rules': Permission denied
tp@ahmed-abdelhedi:~$ sudo ls -l /var/lib/suricata/rules/custom.rules
-rw-r--r-- 1 root suricata 0 Sep 18 20:48 /var/lib/suricata/rules/custom.rules
tp@ahmed-abdelhedi:~$
```

30:



31:



32:

itype:0 : Specifies an ICMP Echo Reply message.

icode:0 : Specifies the standard code value for that Echo Reply.

33:

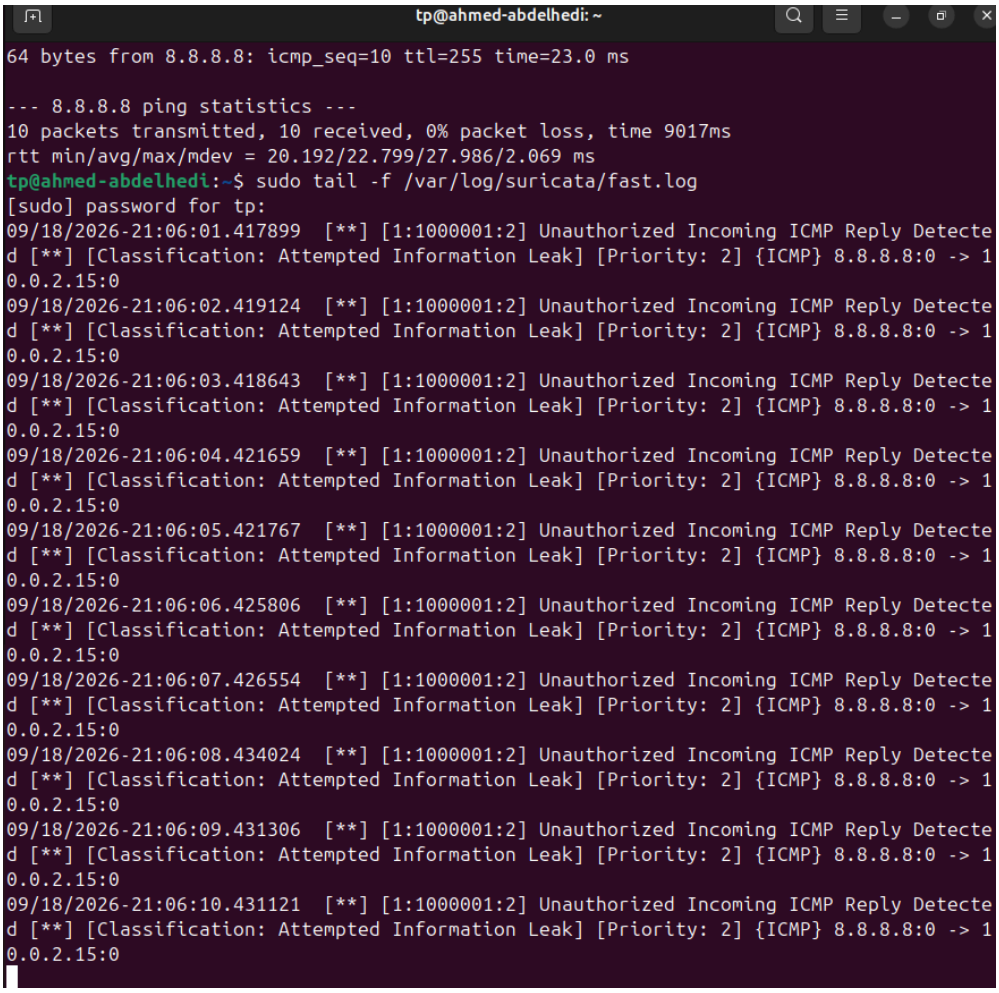
```
tp@ahmed-abdelhedi:~$ sudo suricata -c /etc/suricata/suricata.yaml -v -i enp0s3
Notice: suricata: This is Suricata version 8.0.6 RELEASE running in SYSTEM mode
Info: cpu: CPUs/cores online: 2
Info: suricata: Setting engine mode to IDS mode by default
Info: exception-policy: master exception-policy set to: auto
Info: suricata: Preparing unexpected signal handling
Info: conf: Running in live mode, activating unix socket
Info: logopenfile: fast output device (regular) initialized: fast.log
Info: logopenfile: eve-log output device (regular) initialized: eve.json
Info: logopenfile: stats output device (regular) initialized: stats.log
Info: detect: 2 rule files processed. 52764 rules successfully loaded, 0 rules f
ailed, 0 rules skipped
Info: threshold-config: Threshold config parsed: 0 rule(s) found
Info: detect: 52769 signatures processed. 1227 are IP-only rules, 4534 are inspe
cting packet payload, 46771 inspect application layer, 110 are decoder event onl
y
Info: unix-manager: unix socket '/var/run/suricata/suricata-command.socket'
Info: runmodes: enp0s3: creating 2 threads
Info: ioctl: enp0s3: MTU 1500
Notice: threads: Threads created -> W: 2 FM: 1 FR: 1   Engine started.
```

34:

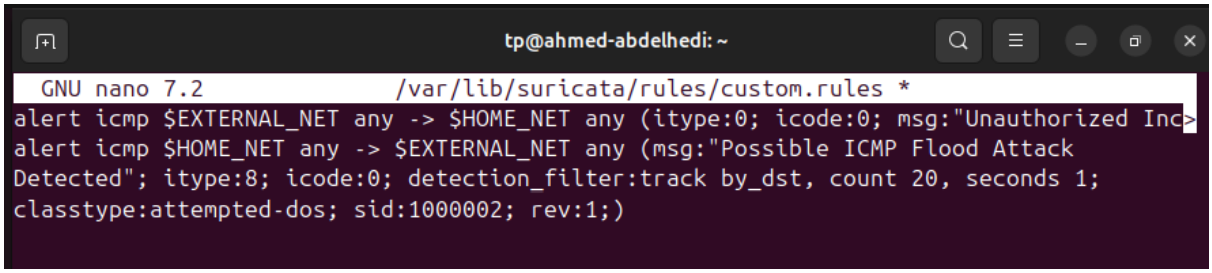
```
tp@ahmed-abdelhedi: ~
tp@ahmed-abdelhedi:~$ ping -c 10 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=255 time=22.0 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=255 time=22.2 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=255 time=20.2 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=255 time=21.9 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=255 time=20.9 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=255 time=22.8 ms
64 bytes from 8.8.8.8: icmp_seq=7 ttl=255 time=22.4 ms
64 bytes from 8.8.8.8: icmp_seq=8 ttl=255 time=28.0 ms
64 bytes from 8.8.8.8: icmp_seq=9 ttl=255 time=24.7 ms
64 bytes from 8.8.8.8: icmp_seq=10 ttl=255 time=23.0 ms

--- 8.8.8.8 ping statistics ---
10 packets transmitted, 10 received, 0% packet loss, time 9017ms
rtt min/avg/max/mdev = 20.192/22.799/27.986/2.069 ms
tp@ahmed-abdelhedi:~$
```

35:



36:



This rule detects a possible ICMP flood when more than 20 Echo Requests are sent to the same destination within one second

37:

```
Info: ioctl: enp0s3: MTU 1500
Notice: threads: Threads created -> W: 2 FM: 1 FR: 1   Engine started.
^CNotice: suricata: Signal Received.  Stopping engine.
Info: suricata: time elapsed 537.580s
Info: counters: Alerts: 10
Notice: device: enp0s3: packets: 106, drops: 0 (0.00%), invalid chksum: 0
tp@ahmed-abdelhedi:~$ sudo suricata -c /etc/suricata/suricata.yaml -v -i enp0s3
Notice: suricata: This is Suricata version 8.0.6 RELEASE running in SYSTEM mode
Info: cpu: CPUs/cores online: 2
Info: suricata: Setting engine mode to IDS mode by default
Info: exception-policy: master exception-policy set to: auto
Info: suricata: Preparing unexpected signal handling
Info: conf: Running in live mode, activating unix socket
Info: logopenfile: fast output device (regular) initialized: fast.log
Info: logopenfile: eve-log output device (regular) initialized: eve.json
Info: logopenfile: stats output device (regular) initialized: stats.log
Info: detect: 2 rule files processed. 52765 rules successfully loaded, 0 rules failed,
0 rules skipped
Info: threshold-config: Threshold config parsed: 0 rule(s) found
Info: detect: 52770 signatures processed. 1227 are IP-only rules, 4534 are inspecting p
acket payload, 46771 inspect application layer, 110 are decoder event only
Info: unix-manager: unix socket '/var/run/suricata/suricata-command.socket'
Info: runmodes: enp0s3: creating 2 threads
Info: ioctl: enp0s3: MTU 1500
Notice: threads: Threads created -> W: 2 FM: 1 FR: 1   Engine started.
```

```
tp@ahmed-abdelhedi: ~
tp@ahmed-abdelhedi:~$ sudo ping -i 0.01 -c 30 198.51.100.1
PING 198.51.100.1 (198.51.100.1) 56(84) bytes of data.

--- 198.51.100.1 ping statistics ---
30 packets transmitted, 0 received, 100% packet loss, time 342ms

tp@ahmed-abdelhedi:~$ sudo tail -f /var/log/suricata/fast.log
09/18/2026-21:18:09.817626  [**] [1:1000002:1] Possible ICMP Flood Attack Detected [**]
 [Classification: Attempted Denial of Service] [Priority: 2] {ICMP} 10.0.2.15:8 -> 198.
51.100.1:0
09/18/2026-21:18:09.829665  [**] [1:1000002:1] Possible ICMP Flood Attack Detected [**]
 [Classification: Attempted Denial of Service] [Priority: 2] {ICMP} 10.0.2.15:8 -> 198.
51.100.1:0
09/18/2026-21:18:09.840913  [**] [1:1000002:1] Possible ICMP Flood Attack Detected [**]
 [Classification: Attempted Denial of Service] [Priority: 2] {ICMP} 10.0.2.15:8 -> 198.
51.100.1:0
09/18/2026-21:18:09.852468  [**] [1:1000002:1] Possible ICMP Flood Attack Detected [**]
 [Classification: Attempted Denial of Service] [Priority: 2] {ICMP} 10.0.2.15:8 -> 198.
51.100.1:0
09/18/2026-21:18:09.864557  [**] [1:1000002:1] Possible ICMP Flood Attack Detected [**]
 [Classification: Attempted Denial of Service] [Priority: 2] {ICMP} 10.0.2.15:8 -> 198.
51.100.1:0
09/18/2026-21:18:09.876698  [**] [1:1000002:1] Possible ICMP Flood Attack Detected [**]
 [Classification: Attempted Denial of Service] [Priority: 2] {ICMP} 10.0.2.15:8 -> 198.
51.100.1:0
09/18/2026-21:18:09.888422  [**] [1:1000002:1] Possible ICMP Flood Attack Detected [**]
 [Classification: Attempted Denial of Service] [Priority: 2] {ICMP} 10.0.2.15:8 -> 198.
51.100.1:0
09/18/2026-21:18:09.899961  [**] [1:1000002:1] Possible ICMP Flood Attack Detected [**]
 [Classification: Attempted Denial of Service] [Priority: 2] {ICMP} 10.0.2.15:8 -> 198.
51.100.1:0
09/18/2026-21:18:09.912406  [**] [1:1000002:1] Possible ICMP Flood Attack Detected [**]
 [Classification: Attempted Denial of Service] [Priority: 2] {ICMP} 10.0.2.15:8 -> 198.
51.100.1:0
09/18/2026-21:18:09.924539  [**] [1:1000002:1] Possible ICMP Flood Attack Detected [**]
 [Classification: Attempted Denial of Service] [Priority: 2] {ICMP} 10.0.2.15:8 -> 198.
51.100.1:0
```

38:

Threshold-based detection can generate false positives during legitimate traffic bursts and may miss slow or distributed attacks. An attacker could remain below the configured threshold or use other types of traffic to perform a DoS attack.

39:

```
tp@ahmed-abdelhedi:~$ cd Downloads/
tp@ahmed-abdelhedi:~/Downloads$ ls
Get_Started_With_Smallpdf.pdf  url.pdf
tp@ahmed-abdelhedi:~/Downloads$ xxd -l 16 Get_Started_With_Smallpdf.pdf
00000000: 2550 4446 2d31 2e37 0d25 e2e3 cfd3 0d0a  %PDF-1.7.%......
tp@ahmed-abdelhedi:~/Downloads$
```

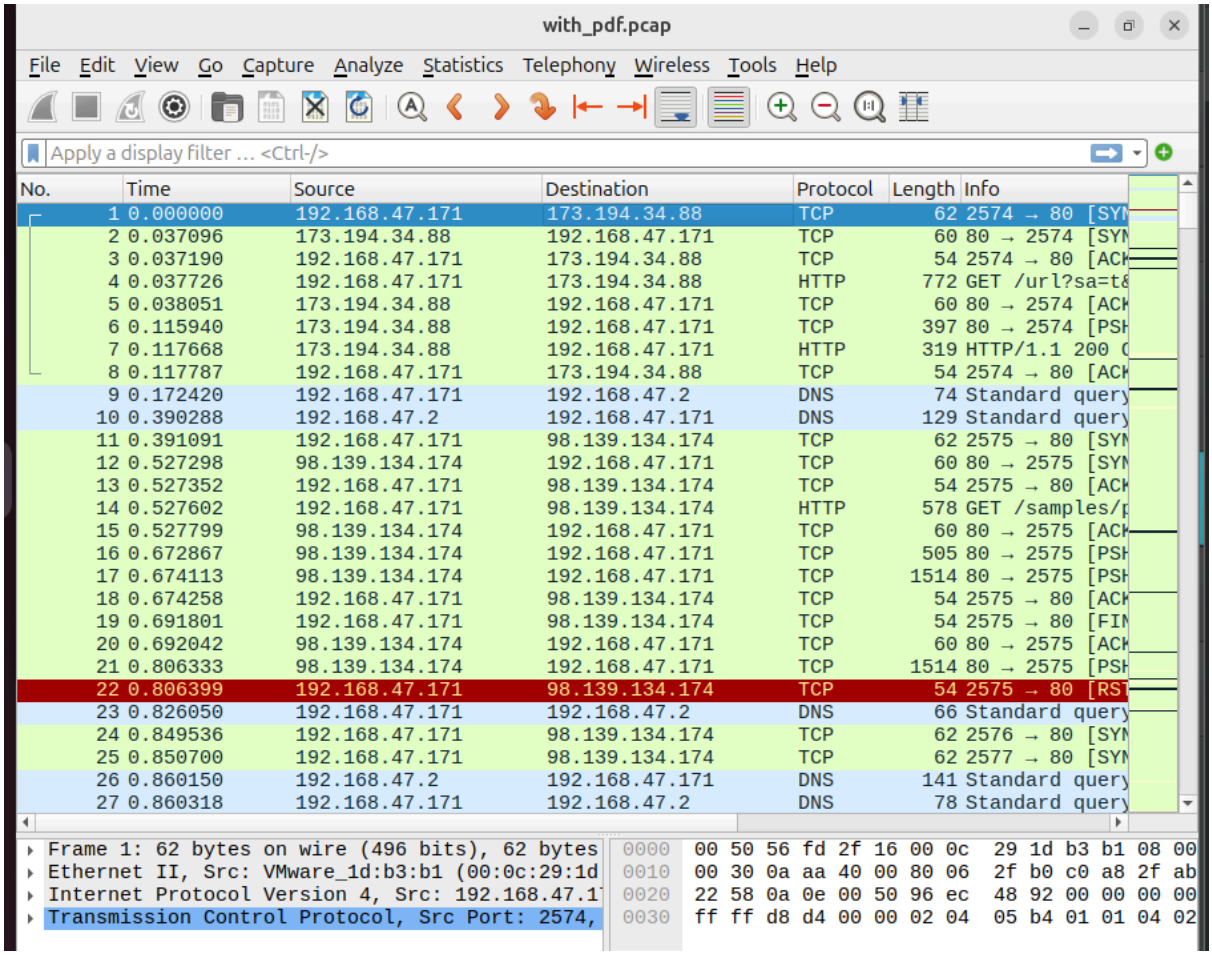
Since Bless was not available, xxd was used to verify that the PDF starts with the magic number 25 50 44 46 2D (%PDF-).

40:

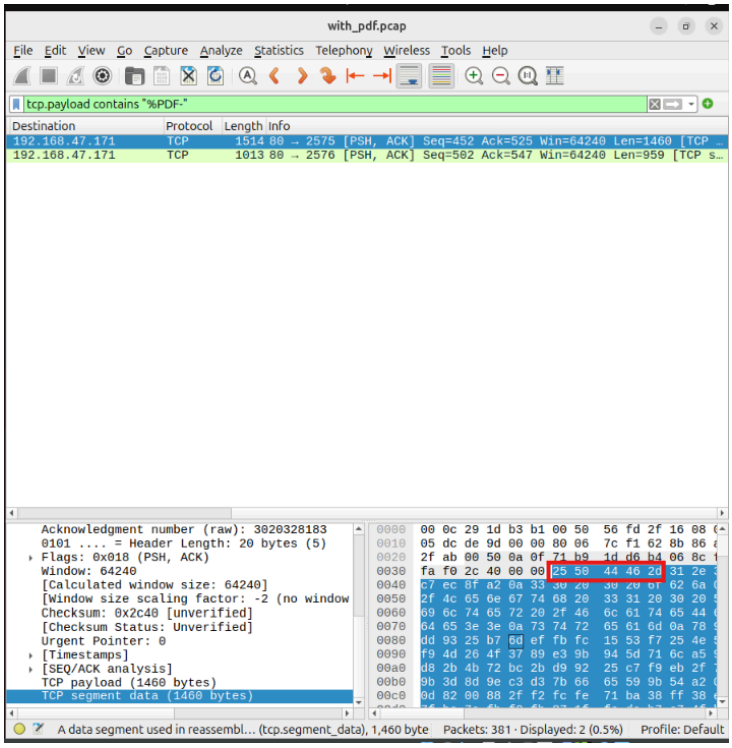
```
tp@ahmed-abdelhedi:~/Downloads$ sudo cat /var/lib/suricata/rules/custom.rules
alert icmp $EXTERNAL_NET any -> $HOME_NET any (itype:0; icode:0; msg:"Unauthorized Inco
ming ICMP Reply Detected"; classtype:attempted-recon; sid:1000001; rev:2;)
alert icmp $HOME_NET any -> $EXTERNAL_NET any (msg:"Possible ICMP Flood Attack Detected
"; itype:8; icode:0; detection_filter:track by_dst, count 20, seconds 1; classtype:atte
mpted-dos; sid:1000002; rev:1;)
alert tcp any any -> $HOME_NET any (msg:"PDF file transfer detected"; content:"|25 50 4
4 46 2D|"; classtype:policy-violation; sid:1000003; rev:1;)
tp@ahmed-abdelhedi:~/Downloads$
```

41:

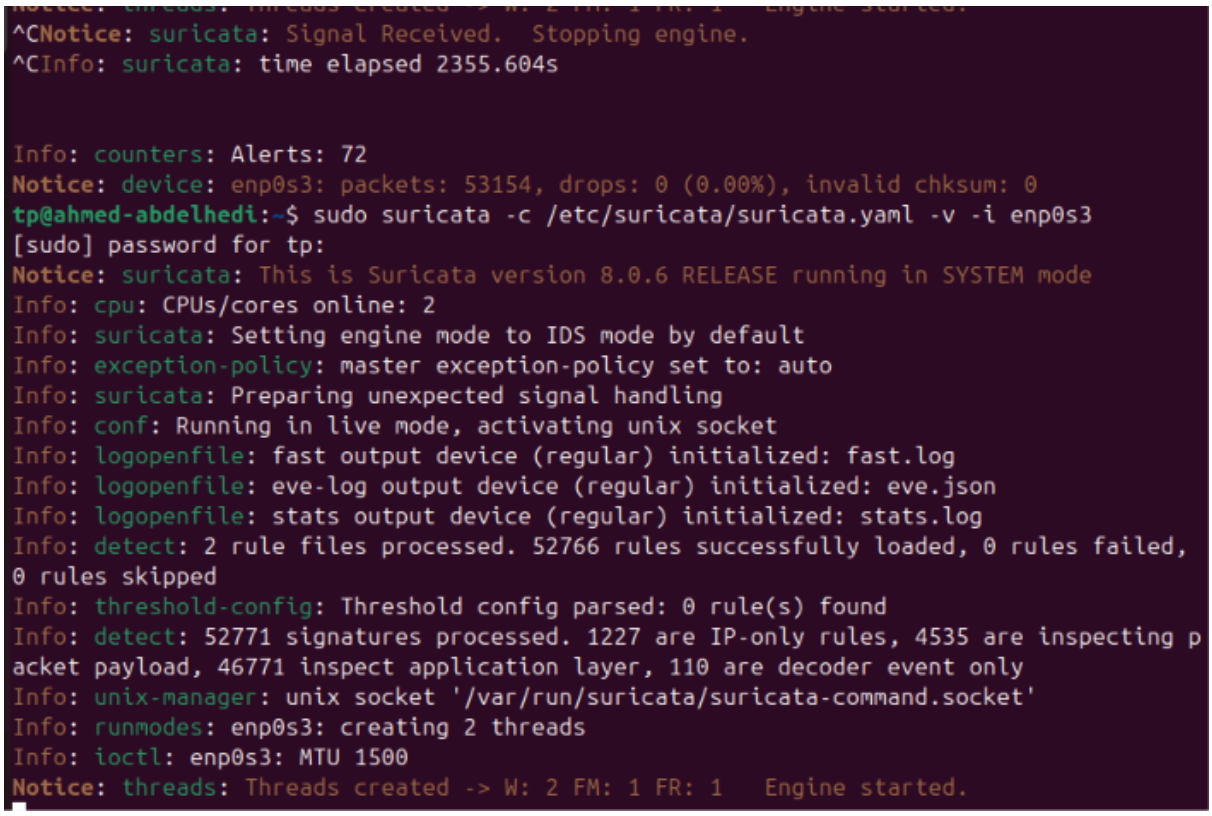
```
tp@ahmed-abdelhedi:~$ ls
Desktop    Downloads  Pictures  snap       test.pdf  with_pdf.pcap
Documents  Music      Public    Templates  Videos    with_pdf.zip
tp@ahmed-abdelhedi:~$
```


with_pdf.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
1 0.000000 192.168.47.171 173.194.34.88 TCP 62 2574 → 80 [SYN
2 0.037096 173.194.34.88 192.168.47.171 TCP 60 80 → 2574 [SYN
3 0.037190 192.168.47.171 173.194.34.88 TCP 54 2574 → 80 [ACK
4 0.037726 192.168.47.171 173.194.34.88 HTTP 772 GET /url?sa=t&
5 0.038051 173.194.34.88 192.168.47.171 TCP 60 80 → 2574 [ACK
6 0.115940 173.194.34.88 192.168.47.171 TCP 397 80 → 2574 [PSH
7 0.117668 173.194.34.88 192.168.47.171 HTTP 319 HTTP/1.1 200 O
8 0.117787 192.168.47.171 173.194.34.88 TCP 54 2574 → 80 [ACK
9 0.172420 192.168.47.171 192.168.47.2 DNS 74 Standard query
10 0.390288 192.168.47.2 192.168.47.171 DNS 129 Standard query
11 0.391091 192.168.47.171 98.139.134.174 TCP 62 2575 → 80 [SYN
12 0.527298 98.139.134.174 192.168.47.171 TCP 60 80 → 2575 [SYN
13 0.527352 192.168.47.171 98.139.134.174 TCP 54 2575 → 80 [ACK
14 0.527602 192.168.47.171 98.139.134.174 HTTP 578 GET /samples/p
15 0.527799 98.139.134.174 192.168.47.171 TCP 60 80 → 2575 [ACK
16 0.672867 98.139.134.174 192.168.47.171 TCP 505 80 → 2575 [PSH
17 0.674113 98.139.134.174 192.168.47.171 TCP 1514 80 → 2575 [PSH
18 0.674258 192.168.47.171 98.139.134.174 TCP 54 2575 → 80 [ACK
19 0.691801 192.168.47.171 98.139.134.174 TCP 54 2575 → 80 [FIN
20 0.692042 98.139.134.174 192.168.47.171 TCP 60 80 → 2575 [ACK
21 0.806333 98.139.134.174 192.168.47.171 TCP 1514 80 → 2575 [PSH
22 0.806399 192.168.47.171 98.139.134.174 TCP 54 2575 → 80 [RST
23 0.826050 192.168.47.171 192.168.47.2 DNS 66 Standard query
24 0.849536 192.168.47.171 98.139.134.174 TCP 62 2576 → 80 [SYN
25 0.850700 192.168.47.171 98.139.134.174 TCP 62 2577 → 80 [SYN
26 0.860150 192.168.47.2 192.168.47.171 DNS 141 Standard query
27 0.860318 192.168.47.171 192.168.47.2 DNS 78 Standard query
Frame 1: 62 bytes on wire (496 bits), 62 bytes
Ethernet II, Src: VMware_1d:b3:b1 (00:0c:29:1d
Internet Protocol Version 4, Src: 192.168.47.1
Transmission Control Protocol, Src Port: 2574,
0000 00 50 56 fd 2f 16 00 0c 29 1d b3 b1 08 00
0010 00 30 0a aa 40 00 80 06 2f b0 c0 a8 2f ab
0020 22 58 0a 0e 00 50 96 ec 48 92 00 00 00 00
0030 ff ff d8 d4 00 00 02 04 05 b4 01 01 04 02

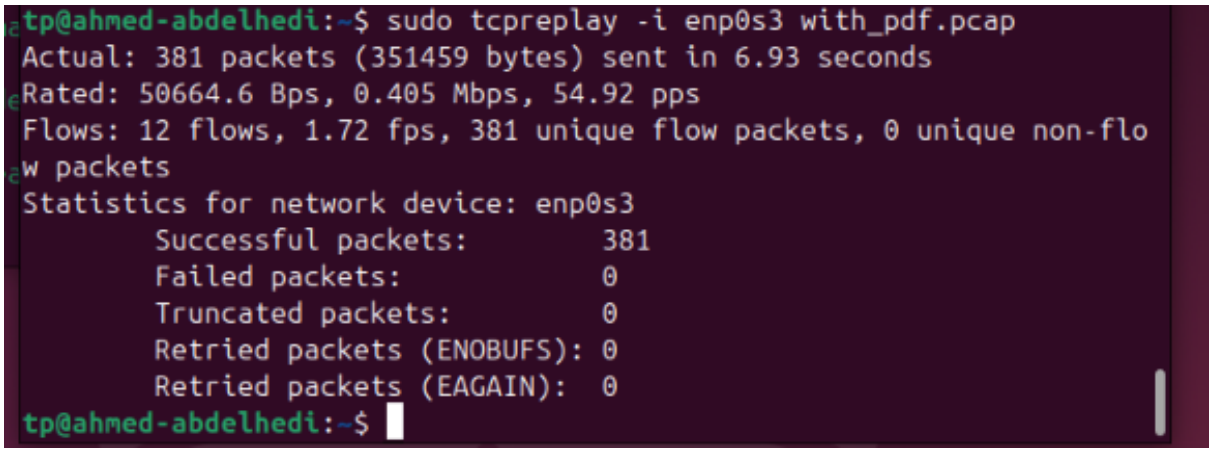
42:


with_pdf.pcap
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
tcp.payload contains "%PDF-"
Destination Protocol Length Info
192.168.47.171 TCP 1514 80 → 2575 [PSH, ACK] Seq=452 Ack=525 Win=64240 Len=1460 [TCP
192.168.47.171 TCP 1013 80 → 2576 [PSH, ACK] Seq=502 Ack=547 Win=64240 Len=959 [TCP s
Acknowledgment number (raw): 3020328183
0101 .... = Header Length: 20 bytes (5)
Flags: 0x018 (PSH, ACK)
Window: 64240
[Calculated window size: 64240]
[Window size scaling factor: -2 (no window
Checksum: 0x2c40 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[SEQ/ACK analysis]
TCP payload (1460 bytes)
TCP segment data (1460 bytes)
A data segment used in reassembl... (tcp.segment_data), 1,460 byte
Packets: 381 · Displayed: 2 (0.5%)
Profile: Default

43:


^CNotice: suricata: Signal Received. Stopping engine.
^CInfo: suricata: time elapsed 2355.604s
Info: counters: Alerts: 72
Notice: device: enp0s3: packets: 53154, drops: 0 (0.00%), invalid chksum: 0
tp@ahmed-abdelhedi:~$ sudo suricata -c /etc/suricata/suricata.yaml -v -i enp0s3
[sudo] password for tp:
Notice: suricata: This is Suricata version 8.0.6 RELEASE running in SYSTEM mode
Info: cpu: CPUs/cores online: 2
Info: suricata: Setting engine mode to IDS mode by default
Info: exception-policy: master exception-policy set to: auto
Info: suricata: Preparing unexpected signal handling
Info: conf: Running in live mode, activating unix socket
Info: logopenfile: fast output device (regular) initialized: fast.log
Info: logopenfile: eve-log output device (regular) initialized: eve.json
Info: logopenfile: stats output device (regular) initialized: stats.log
Info: detect: 2 rule files processed. 52766 rules successfully loaded, 0 rules failed,
0 rules skipped
Info: threshold-config: Threshold config parsed: 0 rule(s) found
Info: detect: 52771 signatures processed. 1227 are IP-only rules, 4535 are inspecting p
acket payload, 46771 inspect application layer, 110 are decoder event only
Info: unix-manager: unix socket '/var/run/suricata/suricata-command.socket'
Info: runmodes: enp0s3: creating 2 threads
Info: ioctl: enp0s3: MTU 1500
Notice: threads: Threads created -> W: 2 FM: 1 FR: 1   Engine started.

44:


tp@ahmed-abdelhedi:~$ sudo tcpreplay -i enp0s3 with_pdf.pcap
Actual: 381 packets (351459 bytes) sent in 6.93 seconds
Rated: 50664.6 Bps, 0.405 Mbps, 54.92 pps
Flows: 12 flows, 1.72 fps, 381 unique flow packets, 0 unique non-flo
w packets
Statistics for network device: enp0s3
Successful packets: 381
Failed packets: 0
Truncated packets: 0
Retried packets (ENOBUFS): 0
Retried packets (EAGAIN): 0
tp@ahmed-abdelhedi:~$

45:

```
tp@ahmed-abdelhedi:~$ sudo grep "PDF file transfer detected" /var/lo
g/suricata/fast.log
09/18/2026-22:02:52.597707  [**] [1:1000003:1] PDF file transfer det
ected [**] [Classification: Potential Corporate Privacy Violation] [
Priority: 1] {TCP} 98.139.134.174:80 -> 192.168.47.171:2575
09/18/2026-22:02:53.293195  [**] [1:1000003:1] PDF file transfer det
ected [**] [Classification: Potential Corporate Privacy Violation] [
Priority: 1] {TCP} 98.139.134.174:80 -> 192.168.47.171:2576
tp@ahmed-abdelhedi:~$
```

46:

Detecting PDF transfers is useful because PDFs can be used to deliver malware or phishing content. However, checking only the %PDF- magic number cannot tell whether the PDF is malicious or safe. Attackers may evade this detection by encrypting, compressing, fragmenting, or modifying the file content so the signature is not easily detected.

V.4. Detecting Embedded JavaScript in PDF Files:

47:

```
tp@ahmed-abdelhedi:~$ sudo nano /var/lib/suricata/rules/custom.rules
[sudo] password for tp:
tp@ahmed-abdelhedi:~$ sudo cat /var/lib/suricata/rules/custom.rules
alert icmp $EXTERNAL_NET any -> $HOME_NET any (itype:0; icode:0; msg:"Unauthorized Inco
ming ICMP Reply Detected"; classtype:attempted-recon; sid:1000001; rev:2;)
alert icmp $HOME_NET any -> $EXTERNAL_NET any (msg:"Possible ICMP Flood Attack Detected
"; itype:8; icode:0; detection_filter:track by_dst, count 20, seconds 1; classtype:atte
mpted-dos; sid:1000002; rev:1;)
alert tcp any any -> $HOME_NET any (msg:"PDF file transfer detected"; content:"|25 50 4
4 46 2D|"; classtype:policy-violation; sid:1000003; rev:1;)
alert http any any -> $HOME_NET any (file.data; content:"|25 50 44 46 2D|"; depth:5; co
ntent:"/JavaScript"; nocase; distance:0; within:8192; flow:to_client,established; msg:"
Suspicious PDF: embedded JavaScript detected"; classtype:policy-violation; sid:1000007;
 rev:2;)
tp@ahmed-abdelhedi:~$
```

This rule detects a PDF file containing embedded JavaScript in HTTP traffic

48:

content:"/JavaScript" → Searches for the /JavaScript keyword inside the PDF.

file.data → Inspects the actual transferred file content.

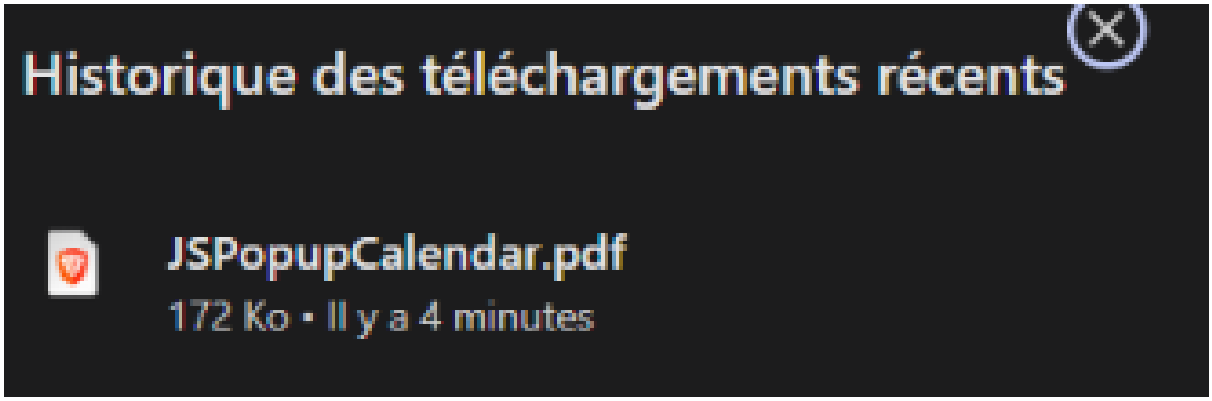
depth:5 → Searches for the PDF signature only within the first 5 bytes.

distance:0 → Starts the second search immediately after the previous matched content.

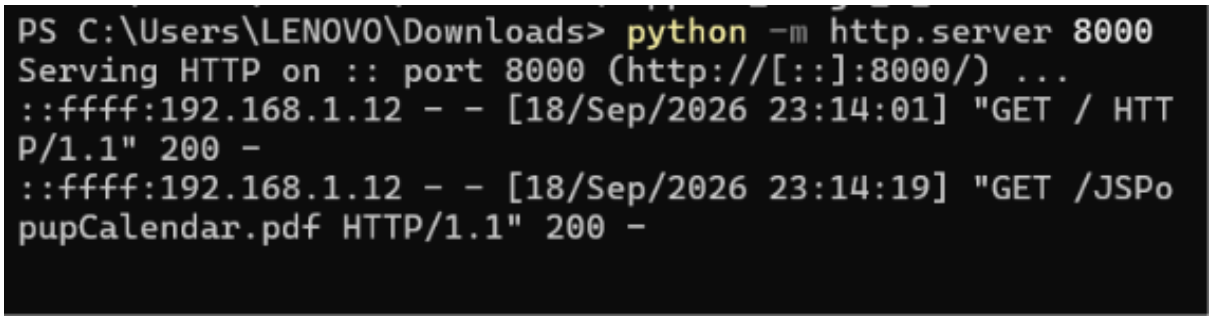
within:8192 → Searches for /JavaScript within the next 8192 bytes.

flow:to_client,established → Applies to an established connection with traffic going toward the client.

49:



50:



51:



52:

```
tp@ahmed-abdelhedi:~$ wget http://192.168.1.12:8000/JSPopupCalendar.pdf
--2026-09-18 23:16:59--  http://192.168.1.12:8000/JSPopupCalendar.pdf
Connecting to 192.168.1.12:8000... connected.
HTTP request sent, awaiting response... 200 OK
Length: 176008 (172K) [application/pdf]
Saving to: 'JSPopupCalendar.pdf.1'

JSPopupCalendar. 100%[==========>] 171.88K  --.-KB/s    in 0.008s

2026-09-18 23:16:59 (21.8 MB/s) - 'JSPopupCalendar.pdf.1' saved [176008/176008]
```

53:

```
tp@ahmed-abdelhedi:~$ sudo tail -f /var/log/suricata/fast.log
09/18/2026-22:01:41.884503  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outboun
d likely related to package management [**] [Classification: Not Suspicious Traffic] [P
riority: 3] {TCP} 10.0.2.15:47094 -> 185.125.190.83:80
09/18/2026-22:01:41.983282  [**] [1:2013504:6] ET INFO GNU/Linux APT User-Agent Outboun
d likely related to package management [**] [Classification: Not Suspicious Traffic] [P
riority: 3] {TCP} 10.0.2.15:47094 -> 185.125.190.83:80
09/18/2026-22:02:52.597707  [**] [1:1000003:1] PDF file transfer detected [**] [Classif
ication: Potential Corporate Privacy Violation] [Priority: 1] {TCP} 98.139.134.174:80 -
> 192.168.47.171:2575
09/18/2026-22:02:53.293195  [**] [1:1000003:1] PDF file transfer detected [**] [Classif
ication: Potential Corporate Privacy Violation] [Priority: 1] {TCP} 98.139.134.174:80 -
> 192.168.47.171:2576
09/18/2026-23:14:00.563378  [**] [1:2034636:2] ET INFO Python SimpleHTTP ServerBanner [
**] [Classification: Misc activity] [Priority: 3] {TCP} 192.168.1.12:8000 -> 10.0.2.15:
57720
09/18/2026-23:14:18.197550  [**] [1:2034636:2] ET INFO Python SimpleHTTP ServerBanner [
**] [Classification: Misc activity] [Priority: 3] {TCP} 192.168.1.12:8000 -> 10.0.2.15:
39960
09/18/2026-23:14:18.198418  [**] [1:1000003:1] PDF file transfer detected [**] [Classif
ication: Potential Corporate Privacy Violation] [Priority: 1] {TCP} 192.168.1.12:8000 -
> 10.0.2.15:39960
09/18/2026-23:16:59.495916  [**] [1:2034636:2] ET INFO Python SimpleHTTP ServerBanner [
**] [Classification: Misc activity] [Priority: 3] {TCP} 192.168.1.12:8000 -> 10.0.2.15:
55646
09/18/2026-23:16:59.496489  [**] [1:1000003:1] PDF file transfer detected [**] [Classif
ication: Potential Corporate Privacy Violation] [Priority: 1] {TCP} 192.168.1.12:8000 -
> 10.0.2.15:55646
09/18/2026-23:16:59.499162  [**] [1:1000007:2] Suspicious PDF: embedded JavaScript dete
cted [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {TCP} 1
92.168.1.12:8000 -> 10.0.2.15:55646
```

V.5. Detecting EXE file transfer

54:

An EXE file can be detected in network traffic by inspecting the content of the transmitted data and looking for its characteristic **file signature (magic number)**.

A Windows Portable Executable (PE) file normally starts with the ASCII characters: MZ

which correspond to the hexadecimal bytes: 4D 5A

Therefore, a Suricata rule can inspect the network payload and search for the MZ signature. If this sequence is found at the beginning of transferred data, it can indicate that an executable file is being transmitted.

However, detecting only MZ is not sufficient to prove that the file is malicious. It only indicates that the traffic may contain a Windows executable.

55:

```
tp@ahmed-abdelhedi:~$ sudo nano /var/lib/suricata/rules/custom.rules
tp@ahmed-abdelhedi:~$ sudo cat /var/lib/suricata/rules/custom.rules
alert icmp $EXTERNAL_NET any -> $HOME_NET any (itype:0; icode:0; msg:"Unauthorized Inco
ming ICMP Reply Detected"; classtype:attempted-recon; sid:1000001; rev:2;)
alert icmp $HOME_NET any -> $EXTERNAL_NET any (msg:"Possible ICMP Flood Attack Detected
"; itype:8; icode:0; detection_filter:track by_dst, count 20, seconds 1; classtype:atte
mpted-dos; sid:1000002; rev:1;)
alert tcp any any -> $HOME_NET any (msg:"PDF file transfer detected"; content:"|25 50 4
4 46 2D|"; classtype:policy-violation; sid:1000003; rev:1;)
alert http any any -> $HOME_NET any (file.data; content:"|25 50 44 46 2D|"; depth:5; co
ntent:"/JavaScript"; nocase; distance:0; within:8192; flow:to_client,established; msg:"
Suspicious PDF: embedded JavaScript detected"; classtype:policy-violation; sid:1000007;
 rev:2;)
alert tcp any any -> $HOME_NET any (msg:"EXE file transfer detected"; content:"|4D 5A|"
; depth:2; classtype:policy-violation; sid:1000004; rev:1;)
tp@ahmed-abdelhedi:~$
```

```
tp@ahmed-abdelhedi: ~
^CInfo: counters: Alerts: 0
Notice: device: enp0s3: packets: 0, drops: 0 (0.00%), invalid chksum: 0
tp@ahmed-abdelhedi:~$ sudo suricata -c /etc/suricata/suricata.yaml -v -i enp0s3
[sudo] password for tp:
Notice: suricata: This is Suricata version 8.0.6 RELEASE running in SYSTEM mode
Info: cpu: CPUs/cores online: 2
Info: suricata: Setting engine mode to IDS mode by default
Info: exception-policy: master exception-policy set to: auto
Info: suricata: Preparing unexpected signal handling
Info: conf: Running in live mode, activating unix socket
Info: logopenfile: fast output device (regular) initialized: fast.log
Info: logopenfile: eve-log output device (regular) initialized: eve.json
Info: logopenfile: stats output device (regular) initialized: stats.log
Info: detect: 2 rule files processed. 52768 rules successfully loaded, 0 rules f
ailed, 0 rules skipped
Info: threshold-config: Threshold config parsed: 0 rule(s) found
Info: detect: 52773 signatures processed. 1227 are IP-only rules, 4536 are inspe
cting packet payload, 46772 inspect application layer, 110 are decoder event onl
y
Info: unix-manager: unix socket '/var/run/suricata/suricata-command.socket'
Info: runmodes: enp0s3: creating 2 threads
Info: ioctl: enp0s3: MTU 1500
Notice: threads: Threads created -> W: 2 FM: 1 FR: 1   Engine started.
```

56:

```
tp@ahmed-abdelhedi:~$ wget http://wildfire.paloaltonetworks.com/publicapi/test/p
e -O test.exe
--2026-09-22 20:27:19--  http://wildfire.paloaltonetworks.com/publicapi/test/pe
Resolving wildfire.paloaltonetworks.com (wildfire.paloaltonetworks.com)... 35.22
2.124.72
Connecting to wildfire.paloaltonetworks.com (wildfire.paloaltonetworks.com)|35.2
22.124.72|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 55296 (54K) [application/octet-stream]
Saving to: ‘test.exe’

test.exe            100%[===================>]  54.00K   139KB/s    in 0.4s

2026-09-22 20:27:20 (139 KB/s) - ‘test.exe’ saved [55296/55296]
```

```
tp@ahmed-abdelhedi:~$ sudo tail -f /var/log/suricata/fast.log
[sudo] password for tp:
09/18/2026-23:14:00.563378  [**] [1:2034636:2] ET INFO Python SimpleHTTP ServerBanner [
**] [Classification: Misc activity] [Priority: 3] {TCP} 192.168.1.12:8000 -> 10.0.2.15:
57720
09/18/2026-23:14:18.197550  [**] [1:2034636:2] ET INFO Python SimpleHTTP ServerBanner [
**] [Classification: Misc activity] [Priority: 3] {TCP} 192.168.1.12:8000 -> 10.0.2.15:
39960
09/18/2026-23:14:18.198418  [**] [1:1000003:1] PDF file transfer detected [**] [Classif
ication: Potential Corporate Privacy Violation] [Priority: 1] {TCP} 192.168.1.12:8000 -
> 10.0.2.15:39960
09/18/2026-23:16:59.495916  [**] [1:2034636:2] ET INFO Python SimpleHTTP ServerBanner [
**] [Classification: Misc activity] [Priority: 3] {TCP} 192.168.1.12:8000 -> 10.0.2.15:
55646
09/18/2026-23:16:59.496489  [**] [1:1000003:1] PDF file transfer detected [**] [Classif
ication: Potential Corporate Privacy Violation] [Priority: 1] {TCP} 192.168.1.12:8000 -
> 10.0.2.15:55646
09/18/2026-23:16:59.499162  [**] [1:1000007:2] Suspicious PDF: embedded JavaScript dete
cted [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {TCP} 1
92.168.1.12:8000 -> 10.0.2.15:55646
09/22/2026-20:27:20.146468  [**] [1:2018959:6] ET INFO PE EXE or DLL Windows file downl
oad HTTP [**] [Classification: Potential Corporate Privacy Violation] [Priority: 1] {TC
P} 35.222.124.72:80 -> 10.0.2.15:46126
09/22/2026-20:27:20.146468  [**] [1:2014520:9] ET INFO EXE - Served Attached HTTP [**]
[Classification: Misc activity] [Priority: 3] {TCP} 35.222.124.72:80 -> 10.0.2.15:46126
09/22/2026-20:27:20.146468  [**] [1:2014819:5] ET INFO Packed Executable Download [**]
[Classification: Misc activity] [Priority: 3] {TCP} 35.222.124.72:80 -> 10.0.2.15:46126
09/22/2026-20:27:20.360376  [**] [1:2015744:6] ET INFO EXE IsDebuggerPresent (Used in M
alware Anti-Debugging) [**] [Classification: Misc activity] [Priority: 3] {TCP} 35.222.
124.72:80 -> 10.0.2.15:46126
^C
tp@ahmed-abdelhedi:~$
```

V.6. Detecting clear text card numbers

57:

```
tp@ahmed-abdelhedi:~$ sudo nano /var/lib/suricata/rules/custom.rules
tp@ahmed-abdelhedi:~$ sudo cat /var/lib/suricata/rules/custom.rules
alert icmp $EXTERNAL_NET any -> $HOME_NET any (itype:0; icode:0; msg:"Unauthorized Inco
ming ICMP Reply Detected"; classtype:attempted-recon; sid:1000001; rev:2;)
alert icmp $HOME_NET any -> $EXTERNAL_NET any (msg:"Possible ICMP Flood Attack Detected
"; itype:8; icode:0; detection_filter:track by_dst, count 20, seconds 1; classtype:atte
mpted-dos; sid:1000002; rev:1;)
alert tcp any any -> $HOME_NET any (msg:"PDF file transfer detected"; content:"|25 50 4
4 46 2D|"; classtype:policy-violation; sid:1000003; rev:1;)
alert http any any -> $HOME_NET any (file.data; content:"|25 50 44 46 2D|"; depth:5; co
ntent:"/JavaScript"; nocase; distance:0; within:8192; flow:to_client,established; msg:"
Suspicious PDF: embedded JavaScript detected"; classtype:policy-violation; sid:1000007;
 rev:2;)
alert tcp any any -> $HOME_NET any (msg:"EXE file transfer detected"; content:"|4D 5A|"
; depth:2; classtype:policy-violation; sid:1000004; rev:1;)
alert tcp $HOME_NET any -> any 25 (flow:to_server,established; smtp.body; pcre:"/4\d{3}
(\s|-)?\d{4}(\s|-)?\d{4}(\s|-)?\d{4}/"; msg:"Sensitive Data Exposure: Possible Visa Car
d Number Detected in Cleartext"; classtype:policy-violation; sid:1000009; rev:1;)
tp@ahmed-abdelhedi:~$
```

58:

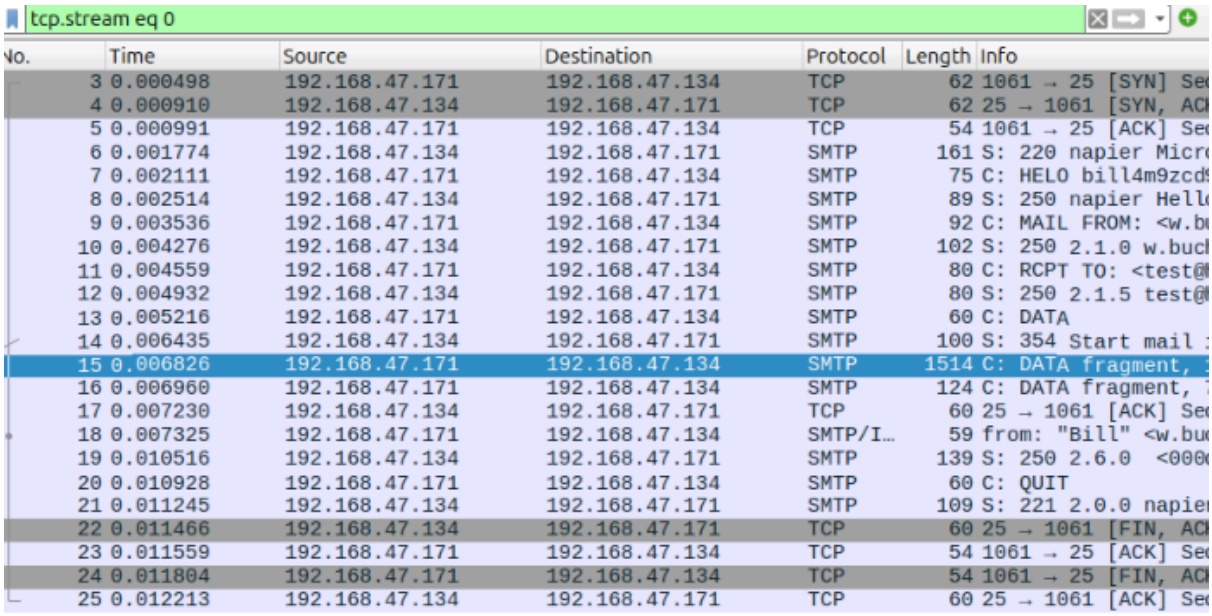
smtp.body : sticky buffer qui restreint la recherche au corps du message SMTP (email), plutôt qu'aux en-têtes ou au paquet entier.

pcre:"/4\d{3}(\s|-)?\d{4}(\s|-)?\d{4}(\s|-)?\d{4}/" : regex correspondant à un numéro Visa 16 chiffres : commence par 4, suivi de 3 chiffres, puis trois groupes de 4 chiffres, chaque groupe séparé optionnellement par un espace ou un tiret

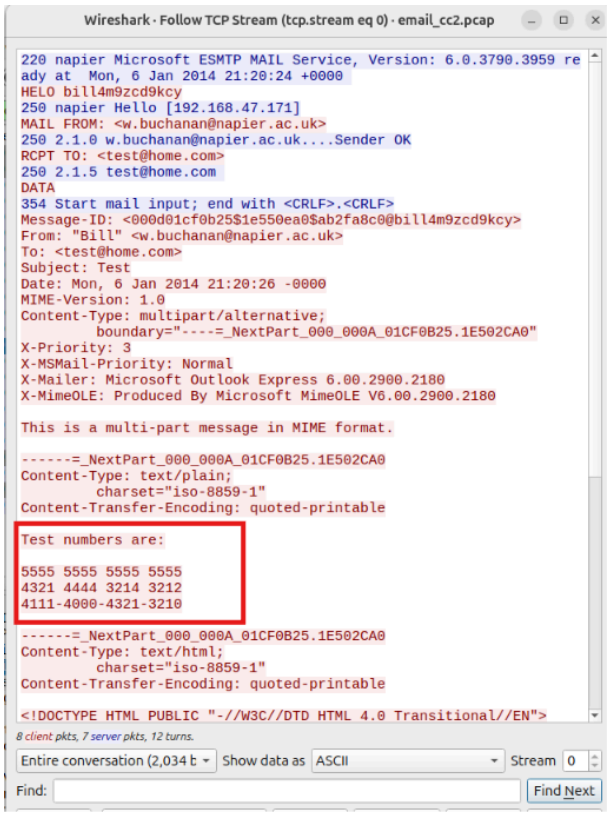
59:

```
tp@ahmed-abdelhedi:~$ ls
Desktop              JSPopupCalendar.pdf.1  snap       wildfire-test-pe-fi
le.exe
Documents            Music                  Templates  with_pdf.pcap
Downloads            pe                     test.exe   with_pdf.zip
email_cc2.zip        Pictures               test.pdf
JSPopupCalendar.pdf  Public                 Videos
tp@ahmed-abdelhedi:~$ unzip email_cc2.zip
Archive:  email_cc2.zip
  inflating: email_cc2.pcap
tp@ahmed-abdelhedi:~$ wireshark email_cc2.pcap
```

60:



the packet contains SMTP traffic



This is the email . Its body is Test numbers are:

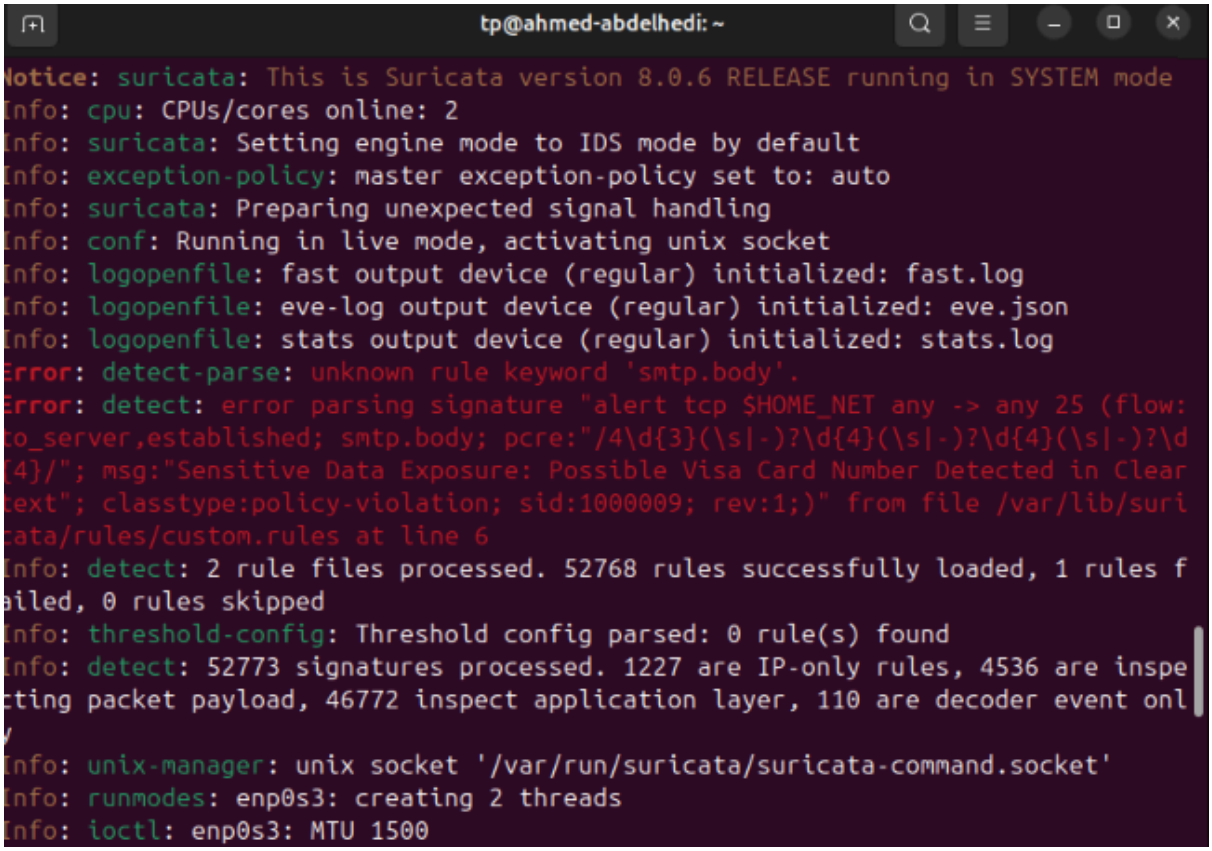
5555 5555 5555 5555

4321 4444 3214 3212

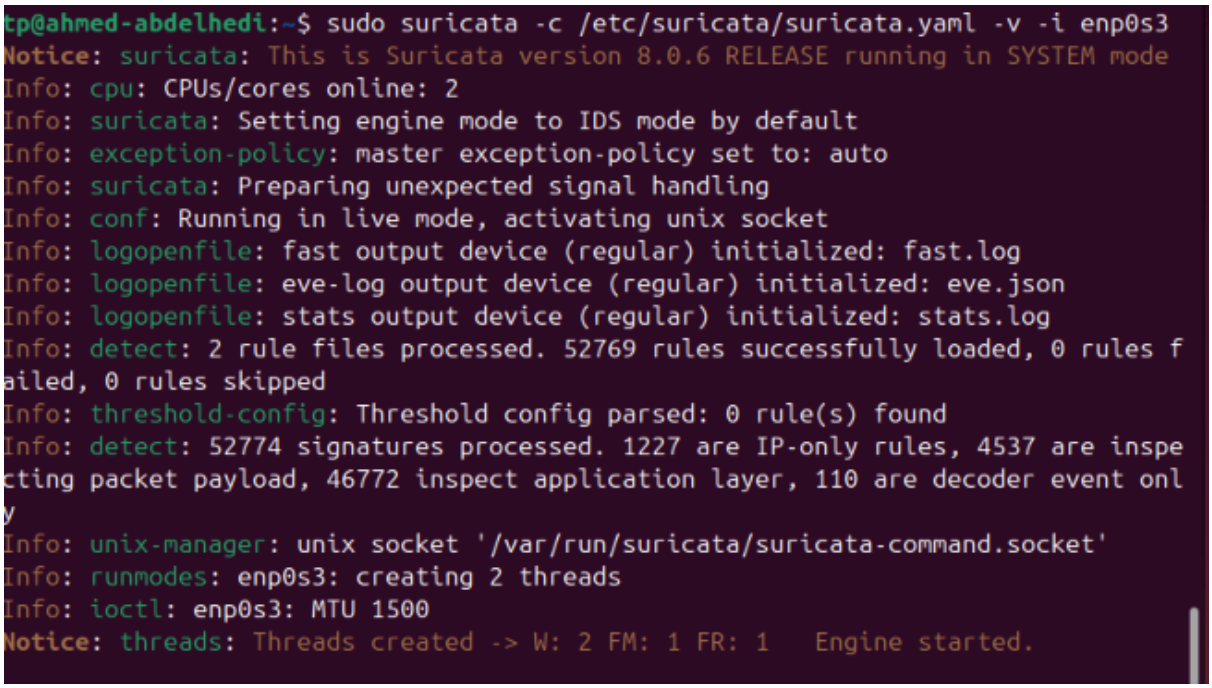
4111-4000-4321-3210

So it very clearly includes a Visa card number

61:



>>> got this error which says the smtp.body keyword is unknown (it might be suricata veersion problem) so i improvised by changing the rule slightly to match the entire TCP payload



62:

```
tp@ahmed-abdelhedi:~$ sudo tcpreplay -i enp0s3 email_cc2.pcap
Actual: 25 packets (3412 bytes) sent in 0.015145 seconds
Rated: 225288.8 Bps, 1.80 Mbps, 1650.70 pps
Flows: 2 flows, 132.05 fps, 23 unique flow packets, 2 unique non-flow pack
ets
Statistics for network device: enp0s3
        Successful packets:        25
        Failed packets:            0
        Truncated packets:         0
        Retried packets (ENOBUFS): 0
        Retried packets (EAGAIN):  0
tp@ahmed-abdelhedi:~$
```

63:

```
tp@ahmed-abdelhedi: ~
tilities for NAT (STUN Binding Request) [**] [Classification: Misc activit
y] [Priority: 3] {UDP} 10.0.2.15:35528 -> 162.159.207.0:3478
09/22/2026-20:44:26.700640  [**] [1:2033078:5] ET INFO Session Traversal U
tilities for NAT (STUN Binding Request On Non-Standard High Port) [**] [Cl
assification: Misc activity] [Priority: 3] {UDP} 10.0.2.15:35528 -> 74.125
.250.129:19302
09/22/2026-20:44:26.707892  [**] [1:2033078:5] ET INFO Session Traversal U
tilities for NAT (STUN Binding Request On Non-Standard High Port) [**] [Cl
assification: Misc activity] [Priority: 3] {UDP} 10.0.2.15:35528 -> 74.125
.250.129:19302
09/22/2026-20:44:26.814971  [**] [1:2016149:4] ET INFO Session Traversal U
tilities for NAT (STUN Binding Request) [**] [Classification: Misc activit
y] [Priority: 3] {UDP} 10.0.2.15:35528 -> 162.159.207.0:3478
09/22/2026-20:44:26.837900  [**] [1:2016150:4] ET INFO Session Traversal U
tilities for NAT (STUN Binding Response) [**] [Classification: Misc activi
ty] [Priority: 3] {UDP} 162.159.207.0:3478 -> 10.0.2.15:35528
09/22/2026-20:57:59.180399  [**] [1:1000009:1] Sensitive Data Exposure: Po
ssible Visa Card Number Detected in Cleartext [**] [Classification: Potent
ial Corporate Privacy Violation] [Priority: 1] {TCP} 192.168.47.171:1061 -
> 192.168.47.134:25
```

64:

Limitations of regex-based detection: It only catches predictable, wellformatted patterns — any deviation (extra spaces, unusual separators, or nonstandard grouping) can slip past, and regex can't verify if a matched number is actually a valid/real card (no checksum validation like Luhn). Evasion: Attackers can encode/obfuscate the data (Base64, encryption/TLS, splitting digits across packets or inserting junk characters), use non-standard

formatting the regex doesn't anticipate, or simply tunnel the traffic over an encrypted channel so the cleartext regex never sees a match.

V.7. Analyzing PCAP file and detecting phishing attacks

65:

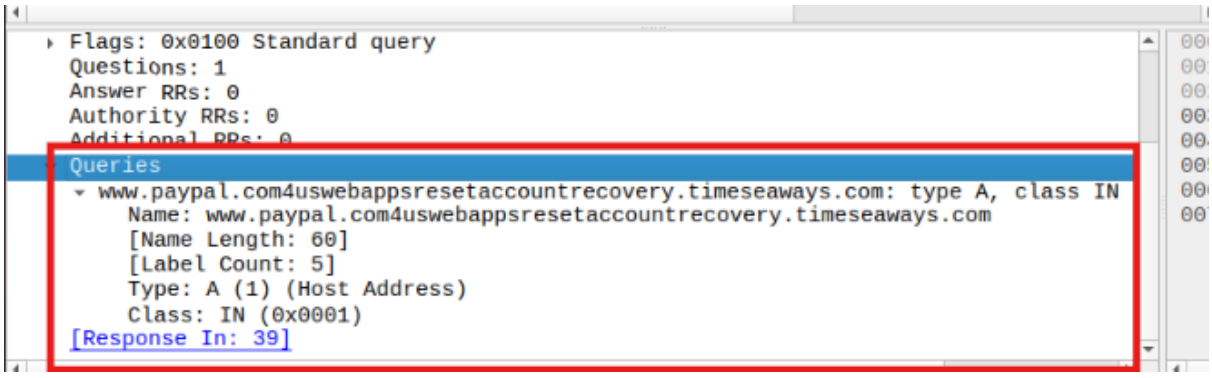
```
tp@ahmed-abdelhedi:~$ wget https://github.com/jishnusaurav/Phishing-attack-PCAP-analysis-using-scapy/raw/master/phishingattack.pcap
--2026-09-22 21:00:45--  https://github.com/jishnusaurav/Phishing-attack-PCAP-analysis-using-scapy/raw/master/phishingattack.pcap
Resolving github.com (github.com)... 140.82.121.4
Connecting to github.com (github.com)|140.82.121.4|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://raw.githubusercontent.com/jishnusaurav/Phishing-attack-PCAP-analysis-using-scapy/master/phishingattack.pcap [following]
--2026-09-22 21:00:46--  https://raw.githubusercontent.com/jishnusaurav/Phishing-attack-PCAP-analysis-using-scapy/master/phishingattack.pcap
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.108.133, 185.199.109.133, 185.199.110.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.108.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 174149 (170K) [application/octet-stream]
Saving to: ‘phishingattack.pcap’

phishingattack.pca 100%[==============>] 170.07K  --.-KB/s    in 0.07s

2026-09-22 21:00:46 (2.23 MB/s) - ‘phishingattack.pcap’ saved [174149/174149]

tp@ahmed-abdelhedi:~$
```

66:



This is a classic phishing/lookalike-domain trick. The real registered domain is timeseaways.com (the attacker's), not PayPal. PayPal’s actual domain is paypal.com which in this case is just stuffed in as a fake subdomain prefix to trick users skimming the URL into thinking it's legitimate. Combined with urgency wording like "resetaccountrecovery," it's a strong phishing indicator.

67/68:

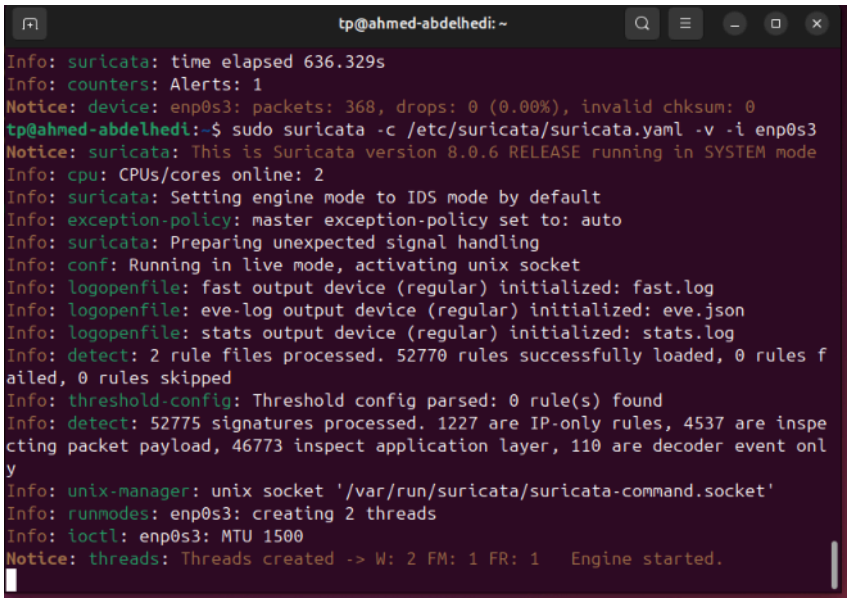
```
tp@ahmed-abdelhedi:~$ sudo nano /var/lib/suricata/rules/custom.rules
tp@ahmed-abdelhedi:~$ sudo cat /var/lib/suricata/rules/custom.rules
alert icmp $EXTERNAL_NET any -> $HOME_NET any (itype:0; icode:0; msg:"Unauthorized Inco
ming ICMP Reply Detected"; classtype:attempted-recon; sid:1000001; rev:2;)
alert icmp $HOME_NET any -> $EXTERNAL_NET any (msg:"Possible ICMP Flood Attack Detected
"; itype:8; icode:0; detection_filter:track by_dst, count 20, seconds 1; classtype:atte
mpted-dos; sid:1000002; rev:1;)
alert tcp any any -> $HOME_NET any (msg:"PDF file transfer detected"; content:"|25 50 4
4 46 2D|"; classtype:policy-violation; sid:1000003; rev:1;)
alert http any any -> $HOME_NET any (file.data; content:"|25 50 44 46 2D|"; depth:5; co
ntent:"/JavaScript"; nocase; distance:0; within:8192; flow:to_client,established; msg:"
Suspicious PDF: embedded JavaScript detected"; classtype:policy-violation; sid:1000007;
 rev:2;)
alert tcp any any -> $HOME_NET any (msg:"EXE file transfer detected"; content:"|4D 5A|"
; depth:2; classtype:policy-violation; sid:1000004; rev:1;)
alert tcp $HOME_NET any -> any 25 (flow:to_server,established; pcre:"/4\d{3}(\s|-)?\d{4
}(\s|-)?\d{4}(\s|-)?\d{4}/"; msg:"Sensitive Data Exposure: Possible Visa Card Number De
tected in Cleartext"; classtype:policy-violation; sid:1000009; rev:1;)
alert dns $HOME_NET any -> $EXTERNAL_NET 53 (msg:"Phishing Attempt: Suspicious PayPal D
omain Detected"; dns_query; content:"paypal.com"; nocase; isdataat:1,relative; classtyp
e:trojan-activity; priority:2; sid:1000010; rev:1;)
tp@ahmed-abdelhedi:~$
```

69:

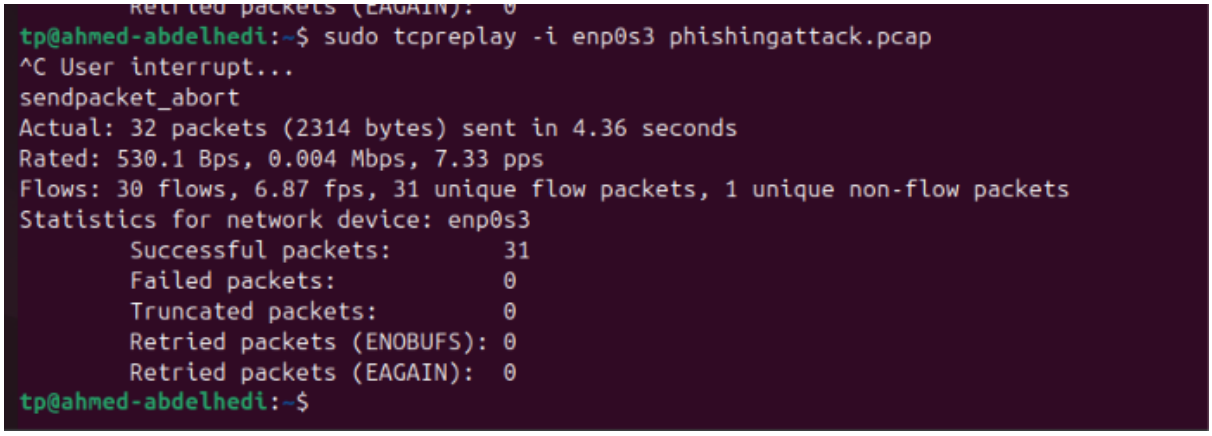
-dns_query — Sticky buffer that restricts the match to the DNS query name field (the domain being looked up), instead of the whole packet.

-isdataat:1,relative — Checks that there's at least 1 more byte of data after the previous match ("paypal.com") within the buffer — i.e., something follows the real domain, which is what flags the spoofed/extended subdomain rather than an exact, clean match on paypal.com alone.

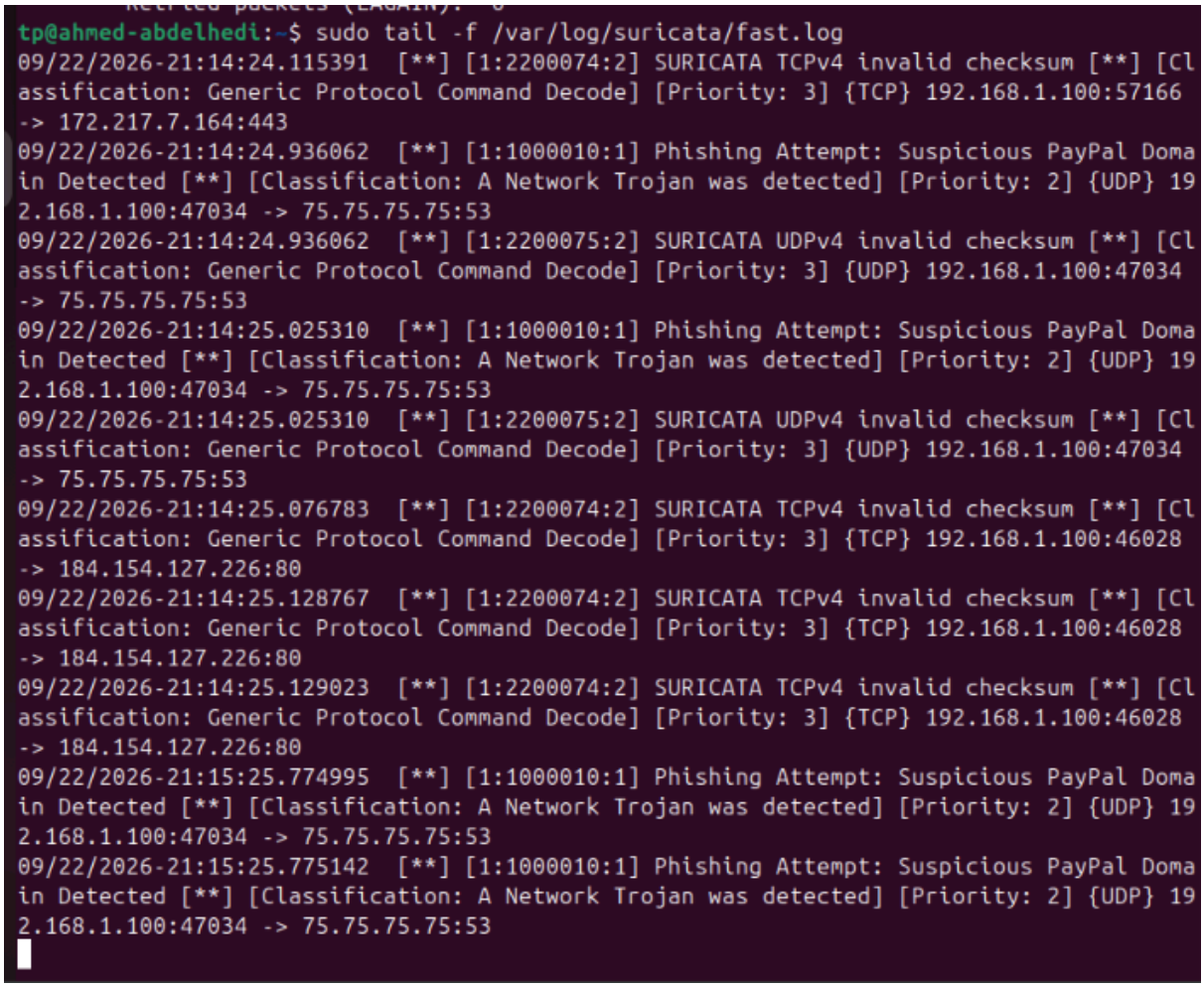
70:


tp@ahmed-abdelhedi: ~
Info: suricata: time elapsed 636.329s
Info: counters: Alerts: 1
Notice: device: enp0s3: packets: 368, drops: 0 (0.00%), invalid chksum: 0
tp@ahmed-abdelhedi:~$ sudo suricata -c /etc/suricata/suricata.yaml -v -i enp0s3
Notice: suricata: This is Suricata version 8.0.6 RELEASE running in SYSTEM mode
Info: cpu: CPUs/cores online: 2
Info: suricata: Setting engine mode to IDS mode by default
Info: exception-policy: master exception-policy set to: auto
Info: suricata: Preparing unexpected signal handling
Info: conf: Running in live mode, activating unix socket
Info: logopenfile: fast output device (regular) initialized: fast.log
Info: logopenfile: eve-log output device (regular) initialized: eve.json
Info: logopenfile: stats output device (regular) initialized: stats.log
Info: detect: 2 rule files processed. 52770 rules successfully loaded, 0 rules failed, 0 rules skipped
Info: threshold-config: Threshold config parsed: 0 rule(s) found
Info: detect: 52775 signatures processed. 1227 are IP-only rules, 4537 are inspecting packet payload, 46773 inspect application layer, 110 are decoder event only
Info: unix-manager: unix socket '/var/run/suricata/suricata-command.socket'
Info: runmodes: enp0s3: creating 2 threads
Info: ioctl: enp0s3: MTU 1500
Notice: threads: Threads created -> W: 2 FM: 1 FR: 1 Engine started.

71:


tp@ahmed-abdelhedi:~$ sudo tcpreplay -i enp0s3 phishingattack.pcap
^C User interrupt...
sendpacket_abort
Actual: 32 packets (2314 bytes) sent in 4.36 seconds
Rated: 530.1 Bps, 0.004 Mbps, 7.33 pps
Flows: 30 flows, 6.87 fps, 31 unique flow packets, 1 unique non-flow packets
Statistics for network device: enp0s3
Successful packets: 31
Failed packets: 0
Truncated packets: 0
Retried packets (ENOBUFS): 0
Retried packets (EAGAIN): 0
tp@ahmed-abdelhedi:~$

72:


tp@ahmed-abdelhedi:~$ sudo tail -f /var/log/suricata/fast.log
09/22/2026-21:14:24.115391 [**] [1:2200074:2] SURICATA TCPv4 invalid checksum [**] [Cl
assification: Generic Protocol Command Decode] [Priority: 3] {TCP} 192.168.1.100:57166
-> 172.217.7.164:443
09/22/2026-21:14:24.936062 [**] [1:1000010:1] Phishing Attempt: Suspicious PayPal Doma
in Detected [**] [Classification: A Network Trojan was detected] [Priority: 2] {UDP} 19
2.168.1.100:47034 -> 75.75.75.75:53
09/22/2026-21:14:24.936062 [**] [1:2200075:2] SURICATA UDPv4 invalid checksum [**] [Cl
assification: Generic Protocol Command Decode] [Priority: 3] {UDP} 192.168.1.100:47034
-> 75.75.75.75:53
09/22/2026-21:14:25.025310 [**] [1:1000010:1] Phishing Attempt: Suspicious PayPal Doma
in Detected [**] [Classification: A Network Trojan was detected] [Priority: 2] {UDP} 19
2.168.1.100:47034 -> 75.75.75.75:53
09/22/2026-21:14:25.025310 [**] [1:2200075:2] SURICATA UDPv4 invalid checksum [**] [Cl
assification: Generic Protocol Command Decode] [Priority: 3] {UDP} 192.168.1.100:47034
-> 75.75.75.75:53
09/22/2026-21:14:25.076783 [**] [1:2200074:2] SURICATA TCPv4 invalid checksum [**] [Cl
assification: Generic Protocol Command Decode] [Priority: 3] {TCP} 192.168.1.100:46028
-> 184.154.127.226:80
09/22/2026-21:14:25.128767 [**] [1:2200074:2] SURICATA TCPv4 invalid checksum [**] [Cl
assification: Generic Protocol Command Decode] [Priority: 3] {TCP} 192.168.1.100:46028
-> 184.154.127.226:80
09/22/2026-21:14:25.129023 [**] [1:2200074:2] SURICATA TCPv4 invalid checksum [**] [Cl
assification: Generic Protocol Command Decode] [Priority: 3] {TCP} 192.168.1.100:46028
-> 184.154.127.226:80
09/22/2026-21:15:25.774995 [**] [1:1000010:1] Phishing Attempt: Suspicious PayPal Doma
in Detected [**] [Classification: A Network Trojan was detected] [Priority: 2] {UDP} 19
2.168.1.100:47034 -> 75.75.75.75:53
09/22/2026-21:15:25.775142 [**] [1:1000010:1] Phishing Attempt: Suspicious PayPal Doma
in Detected [**] [Classification: A Network Trojan was detected] [Priority: 2] {UDP} 19
2.168.1.100:47034 -> 75.75.75.75:53